University of Otago
Te Whare Wānanga o Otāgo
New Zealand

Department of
Computer Science

IVCNZ2020

Predicting Cherry Quality Using Siamese Networks

Yerren van Sint Annaland

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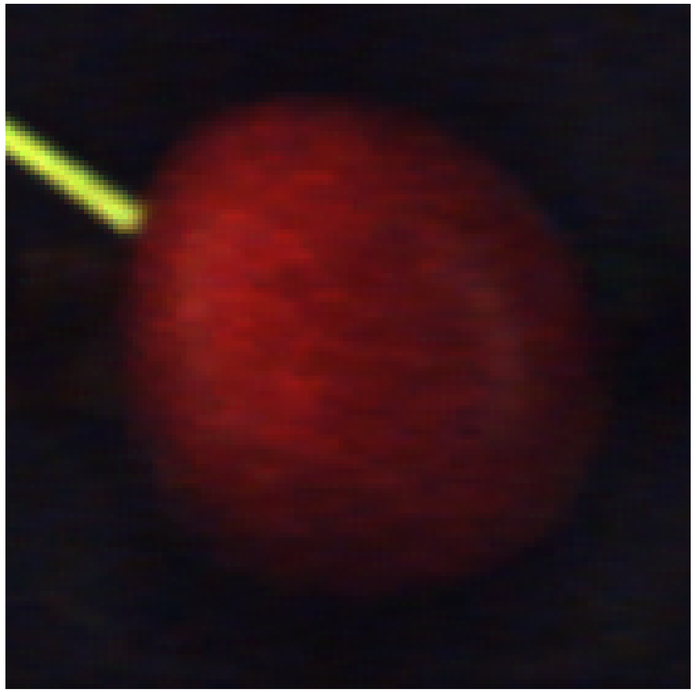
"SeeingRed"

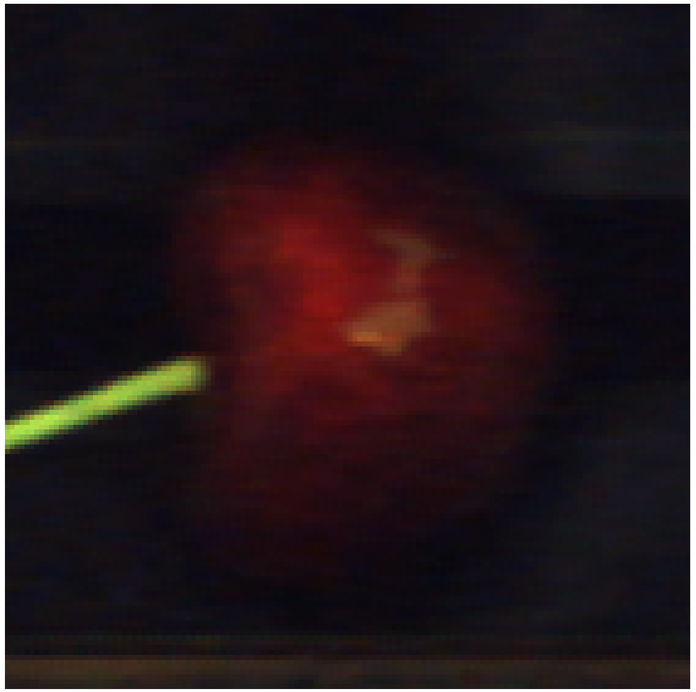
Yerren van Sint Annaland

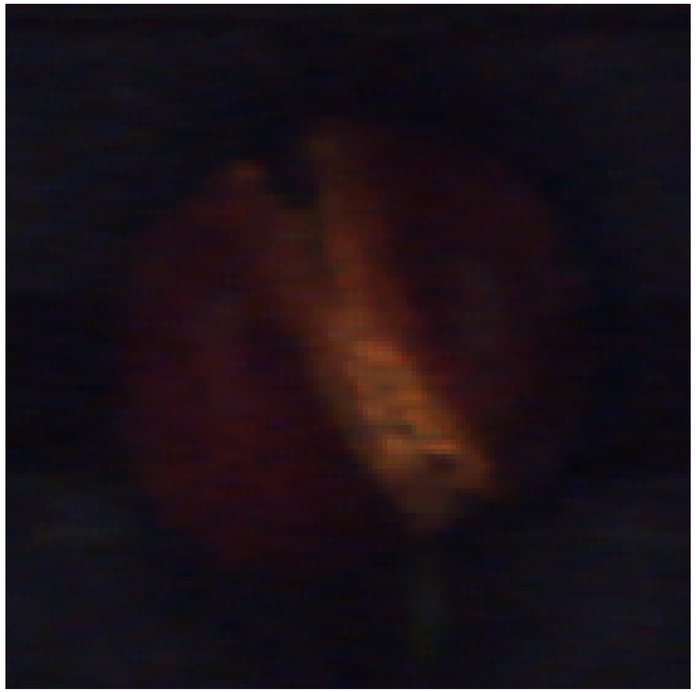
Lech Szymanski

Steven Mills

Grading

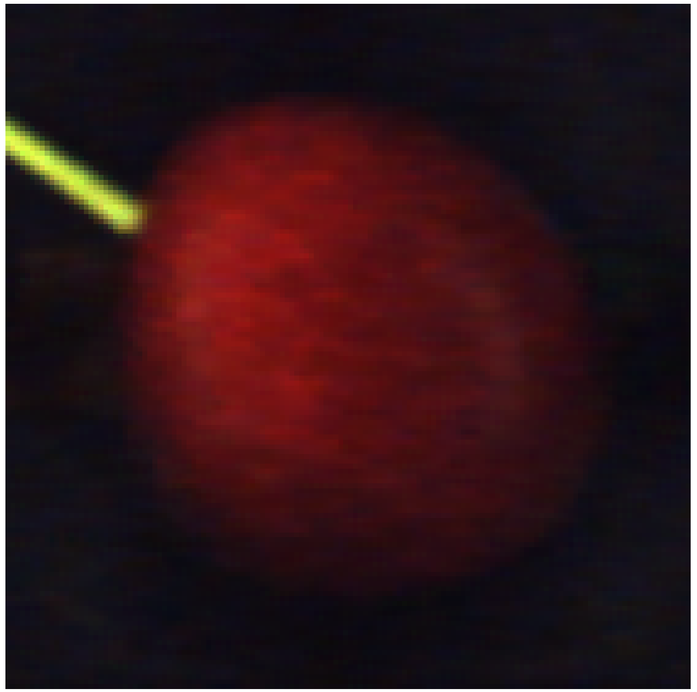




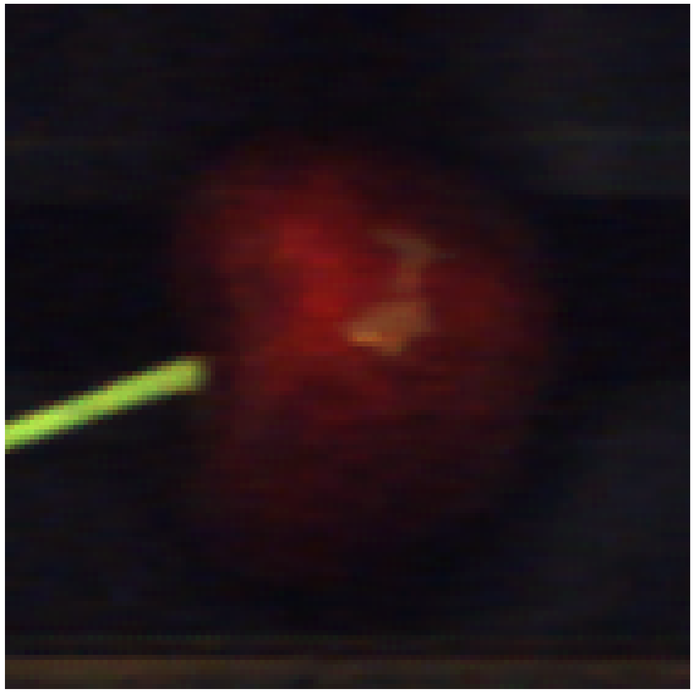


Which one do you think a grader should give you?

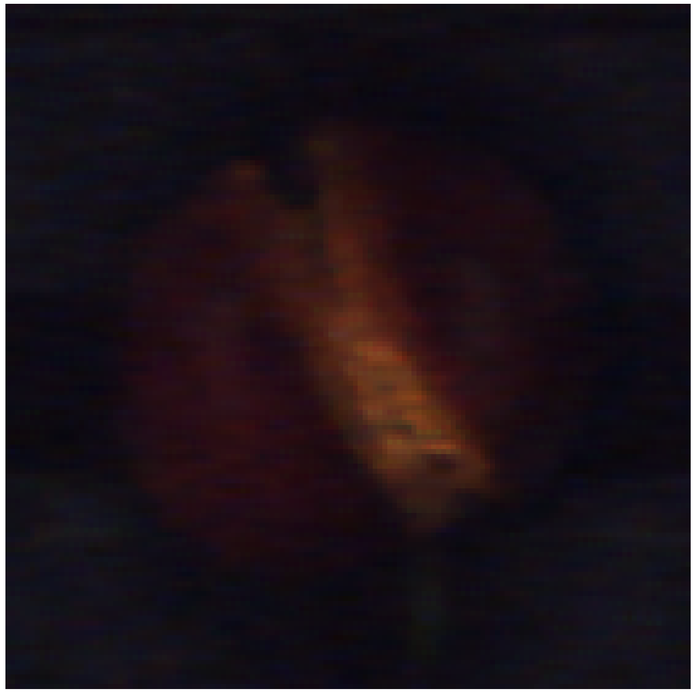
Grading



Export



Local



Cull

Grading

What about this one?



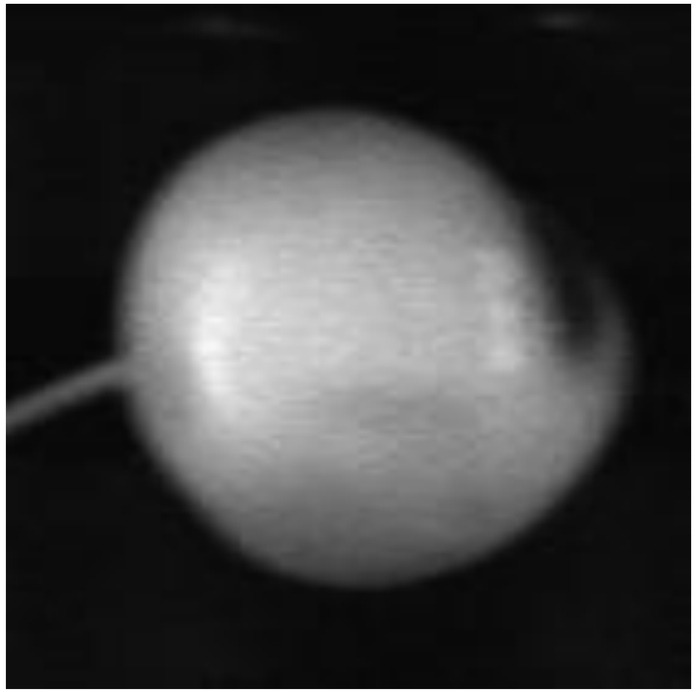
Grading

What about this one?

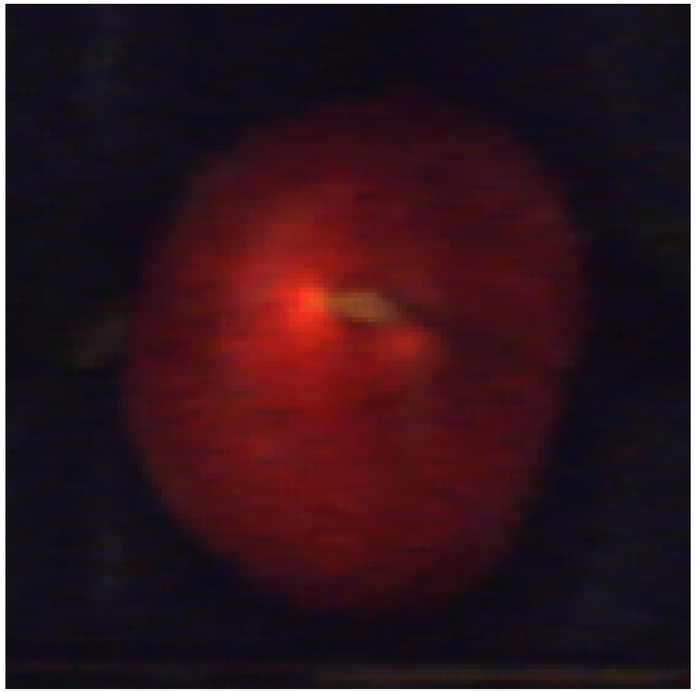
RGB

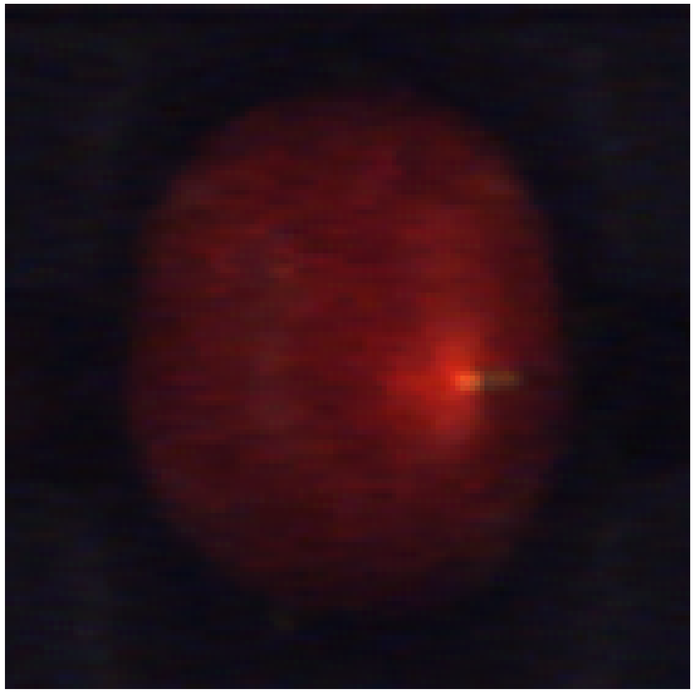


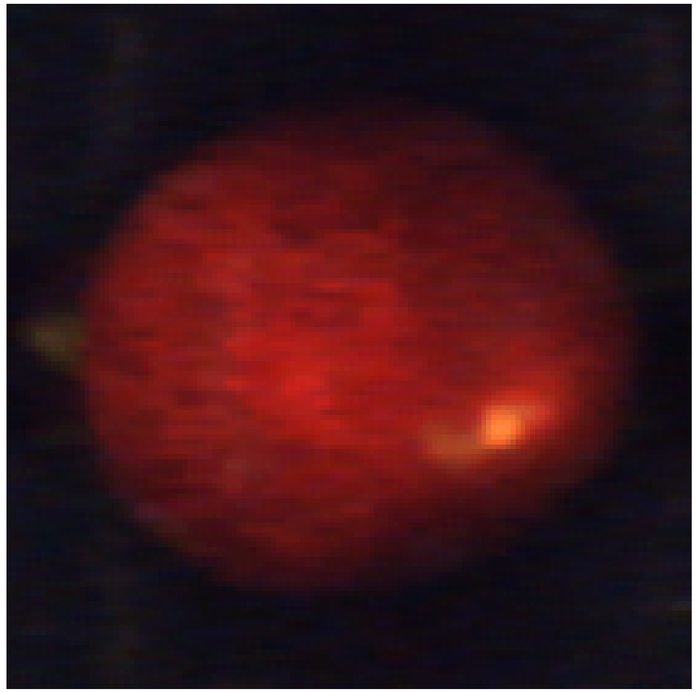
Near Infrared



Grading







It’s not always easy…

What’s More...

The grade standards are constantly changing

The Existing “Solution”

Extracted attributes

- Brightness
- Size
- Colour (Brown)
- …

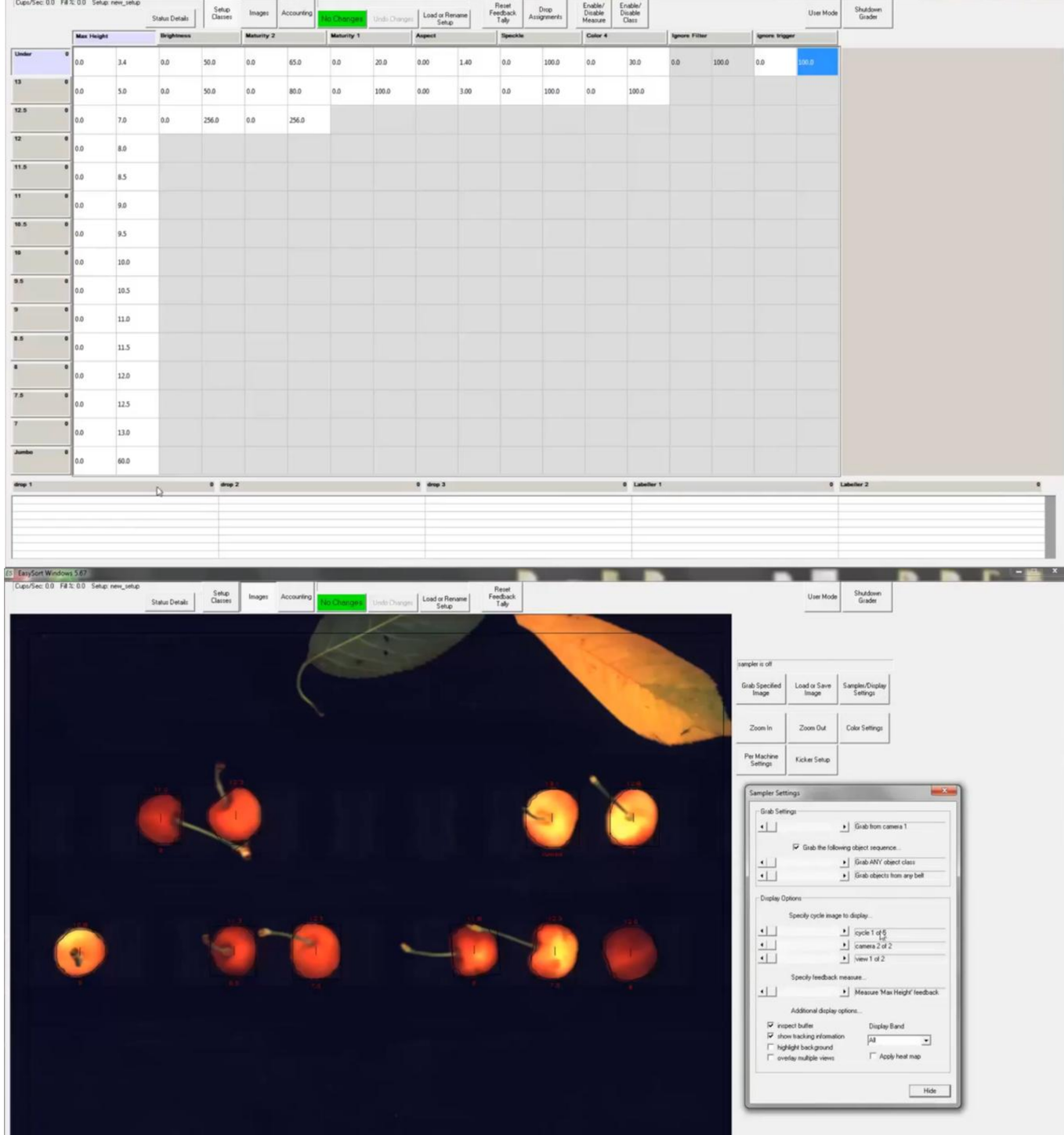


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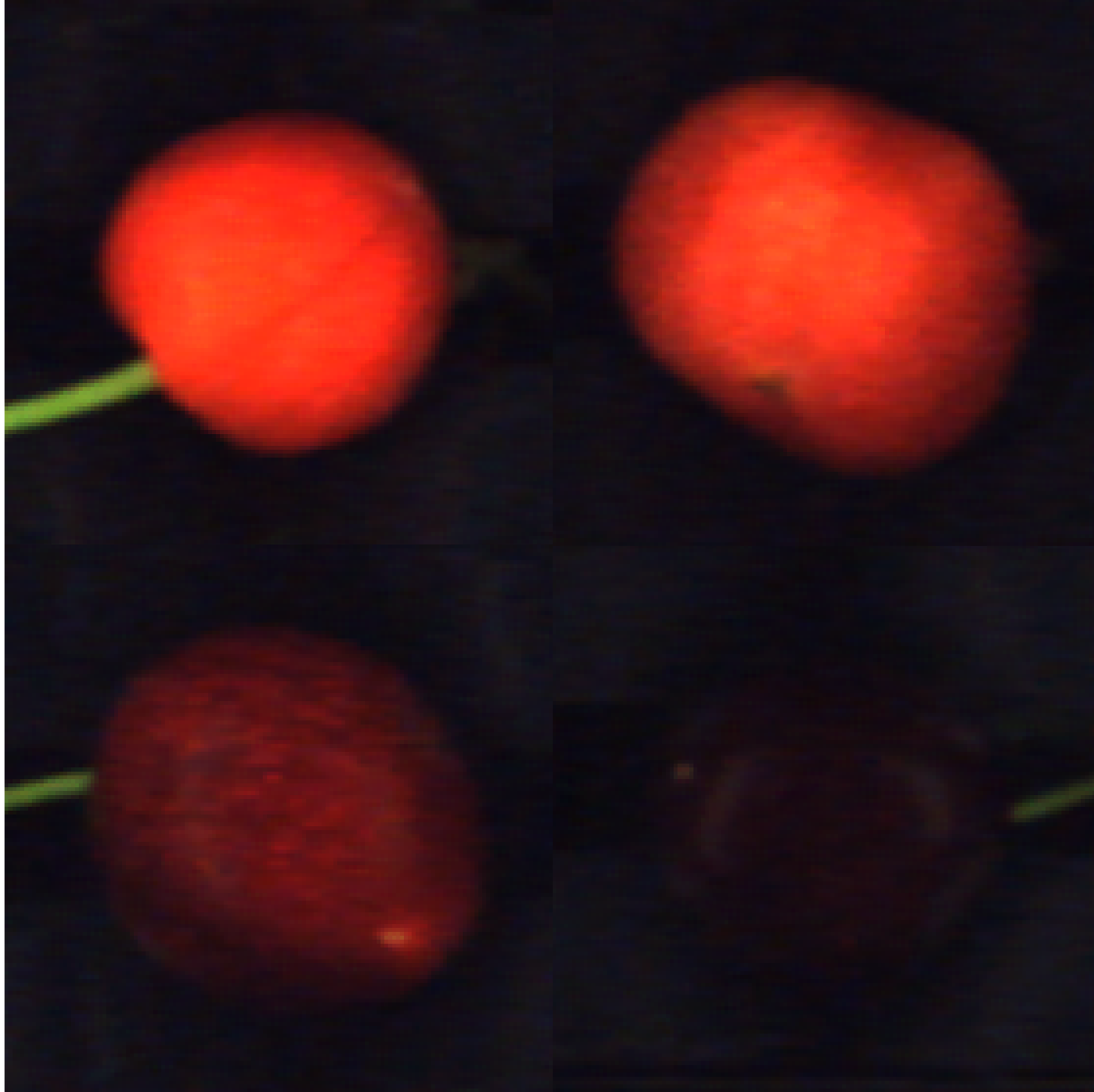




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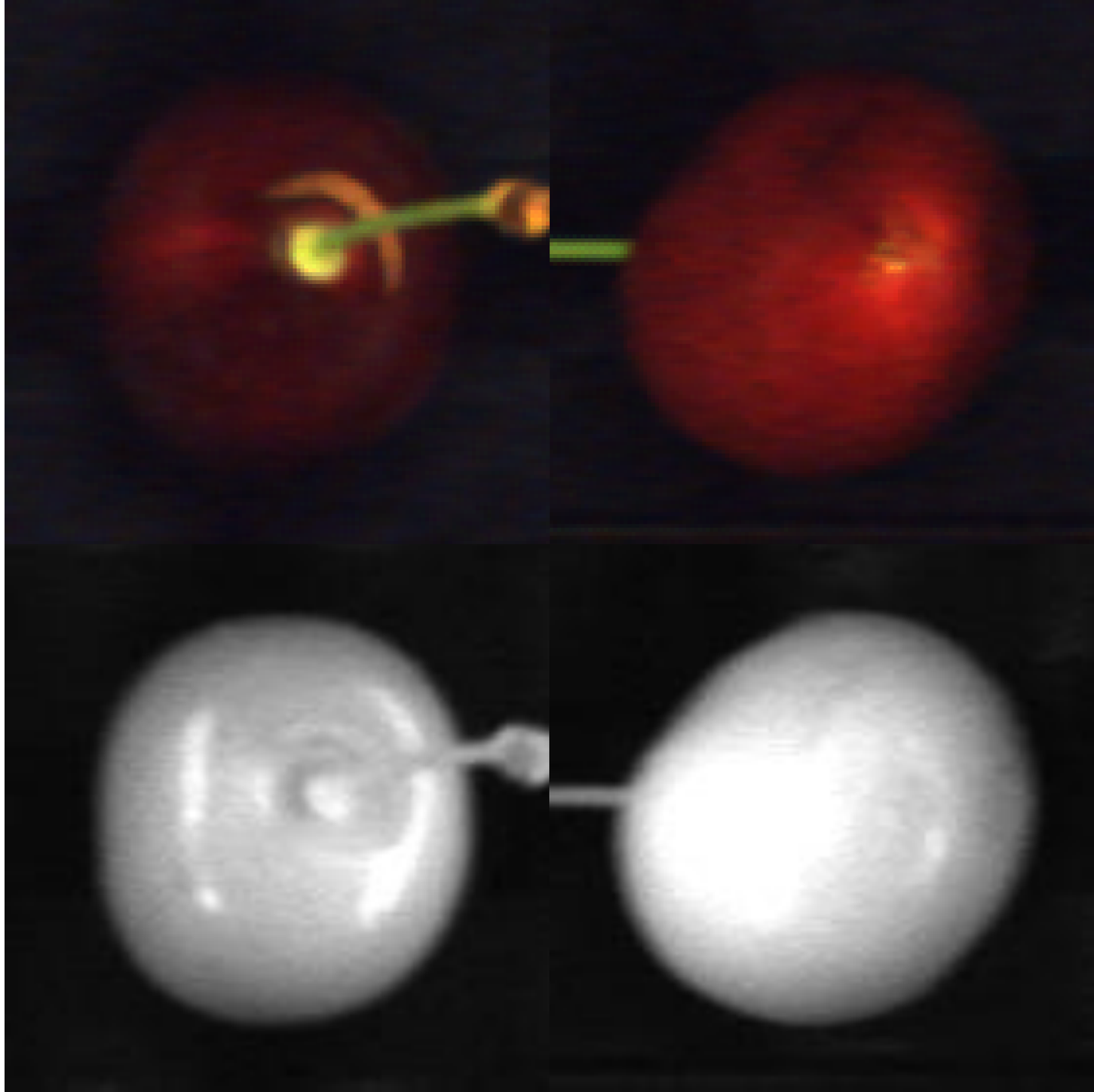
The Existing “Solution”

Extracted attributes

- Brightness
- Size
- Colour (Brown)
- …

Enough for a robust system?

- Colour/Brightness variation
- Often gets confused due to stems
- Cannot spot small defects
- Overlooks obvious rot
- Does not understand cracks seen from the side on
- …



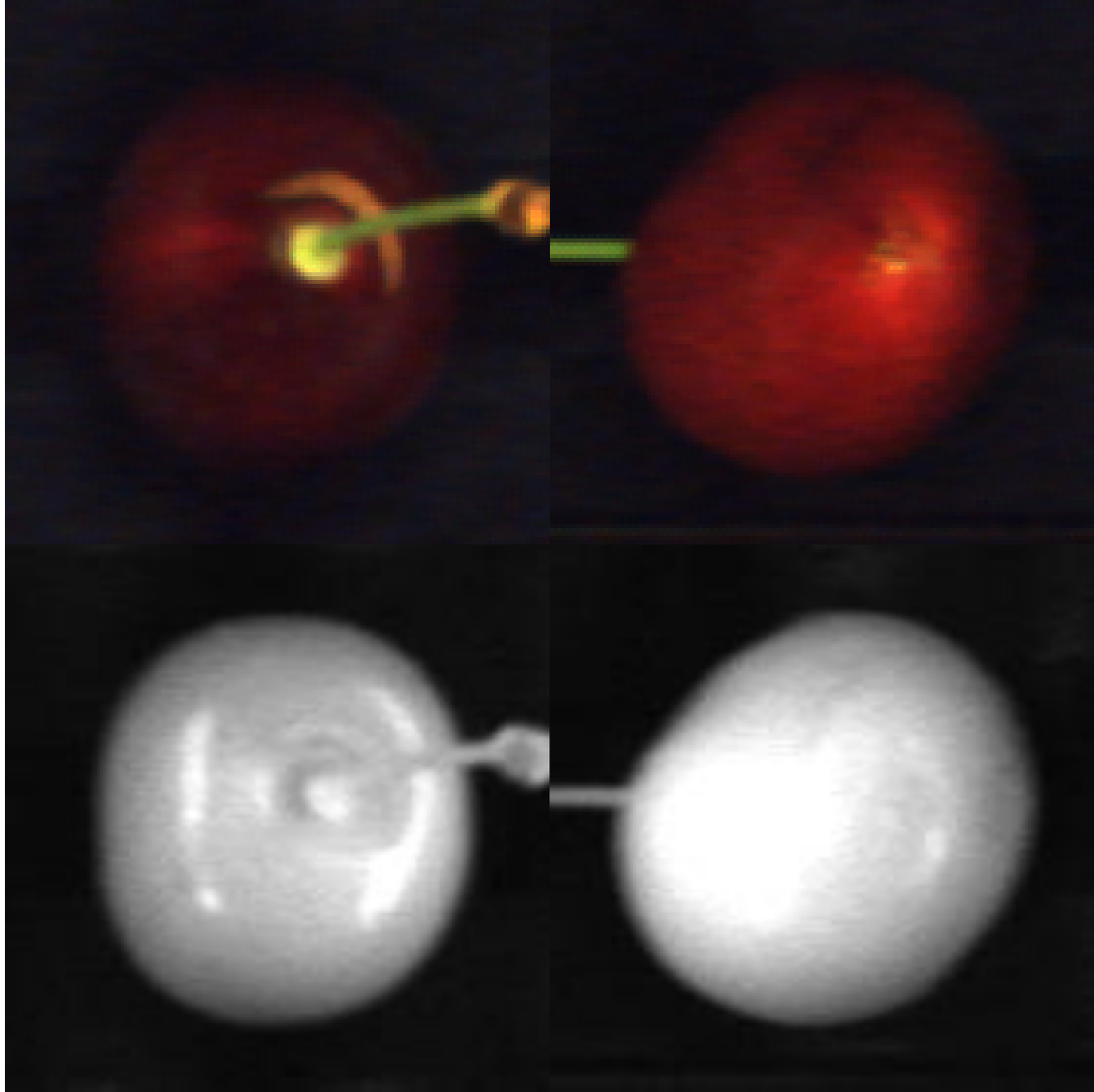
Deep Learning

Fruit grading

- In 2019-2020, Naranjo-Torres et al. [1] found 18 publications where convolutional neural networks (CNNs) were used for fruit imaging tasks.
- Out of these, only five were concerned with quality control.

Cherries

- Momeny et al. used a CNN to classify cherry shapes [2].
- Villacrés and Cheein estimate cherry harvest yields using a CNN [3].



[1] J. Naranjo-Torres, M. Mora, R. Hernández-García, R. J. Barrientos, C. Fredes, and A. Valenzuela, "A Review of Convolutional Neural Network Applied to Fruit Image Processing," *Applied Sciences*, vol. 10, no. 10, p. 3443, Jan. 2020, doi: 10.3390/app10103443.

[2] M. Momeny, A. Jahanbakhshi, K. Jafarnezhad, and Y.-D. Zhang, "Accurate classification of cherry fruit using deep CNN based on hybrid pooling approach," *Postharvest Biology and Technology*, vol. 166, p. 111204, Aug. 2020, doi: 10.1016/j.postharvbio.2020.111204.

[3] J. F. Villacrés and F. Auat Cheein, "Detection and Characterization of Cherries: A Deep Learning Usability Case Study in Chile," *Agronomy*, vol. 10, no. 6, p. 835, Jun. 2020, doi: 10.3390/agronomy10060835.

The Problem

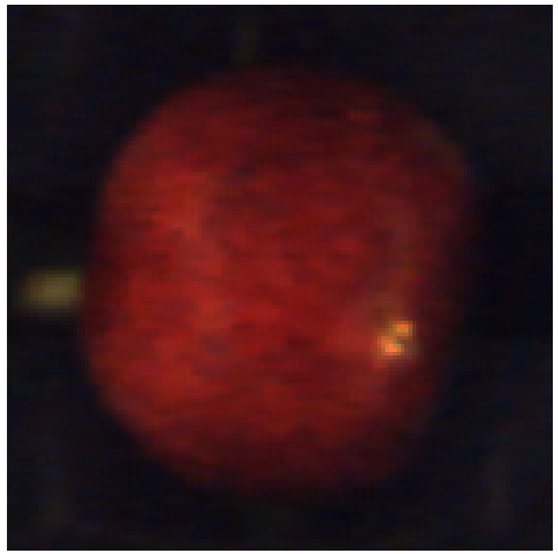
The Real Problem

Flexible grade standards. Again.

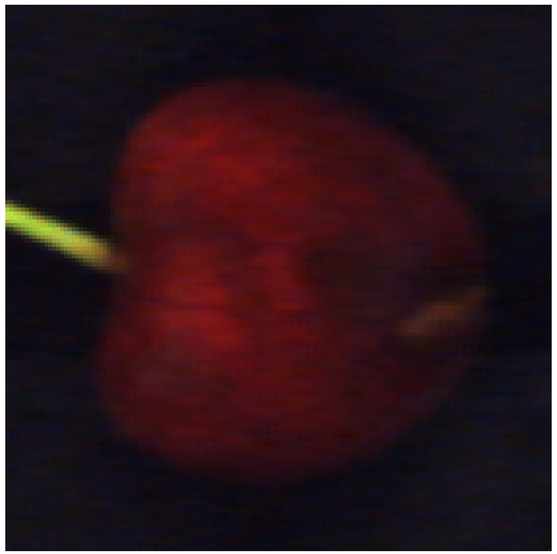
The Real Problem

Cannot use fixed classes

- Grade standards constantly change
- Traditional classification methods unsuitable



Export or local?



Local or cull?

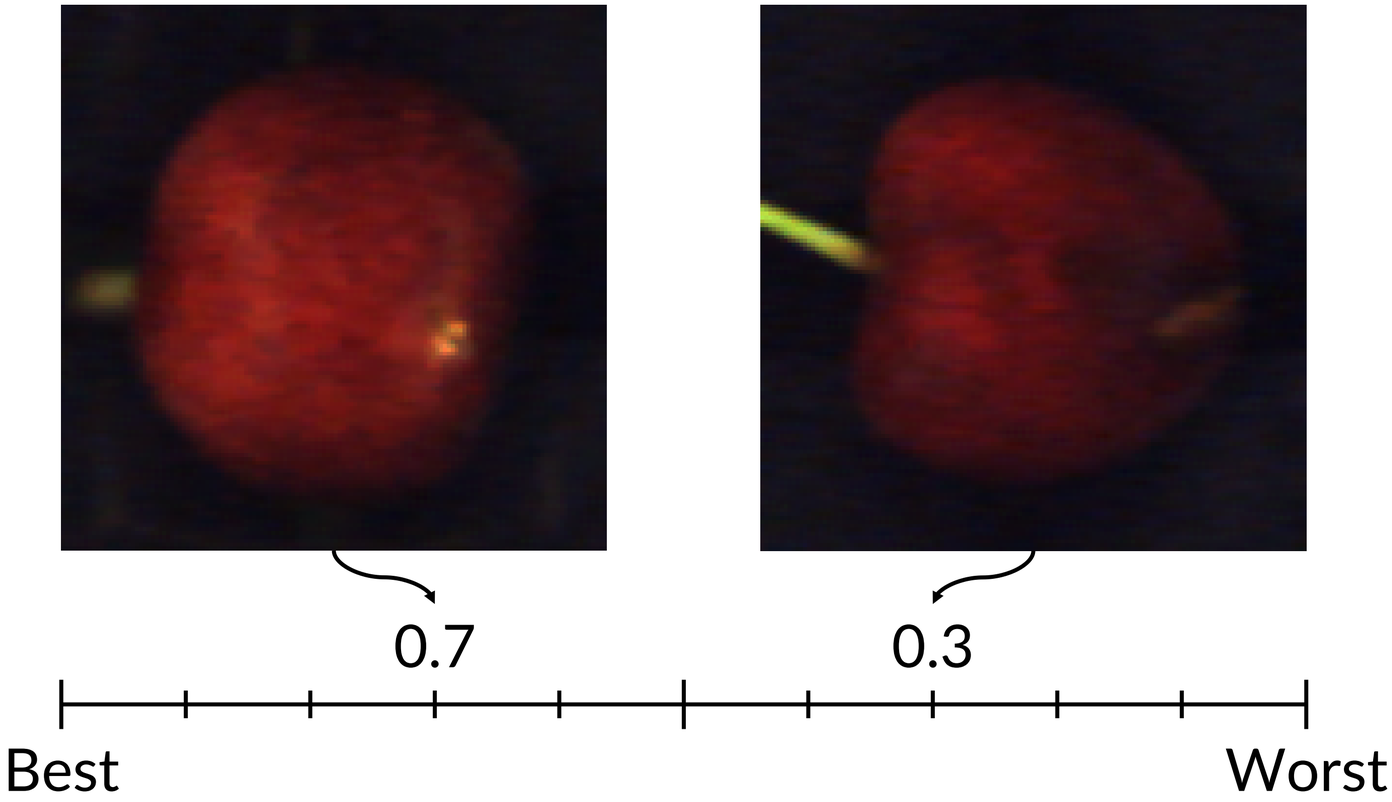
The Real Problem

Cannot use fixed classes

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- Traditional classification methods unsuitable

Continuous scale

- Train a model to predict ‘quality values’
- Just re-label images with quality values…?



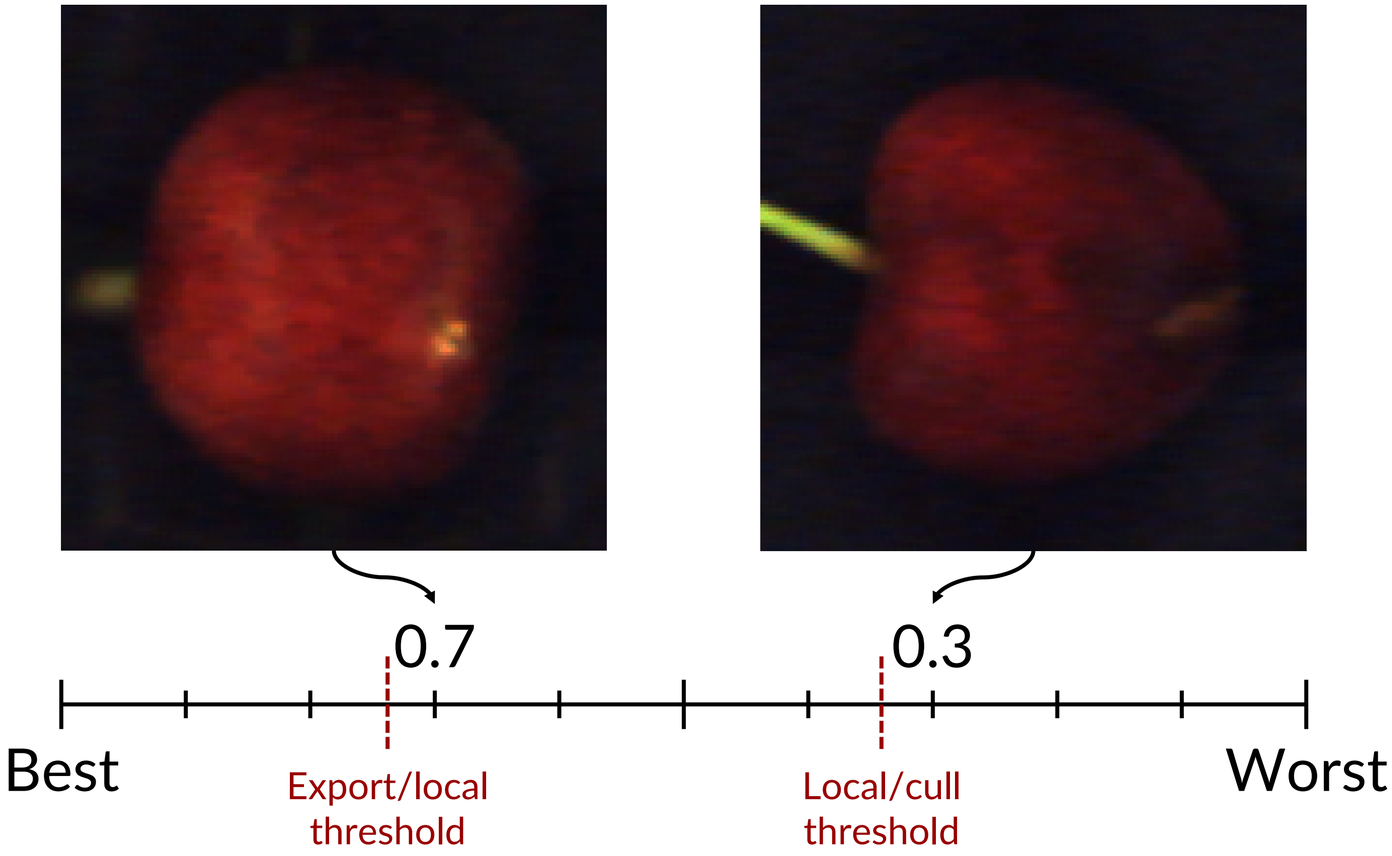
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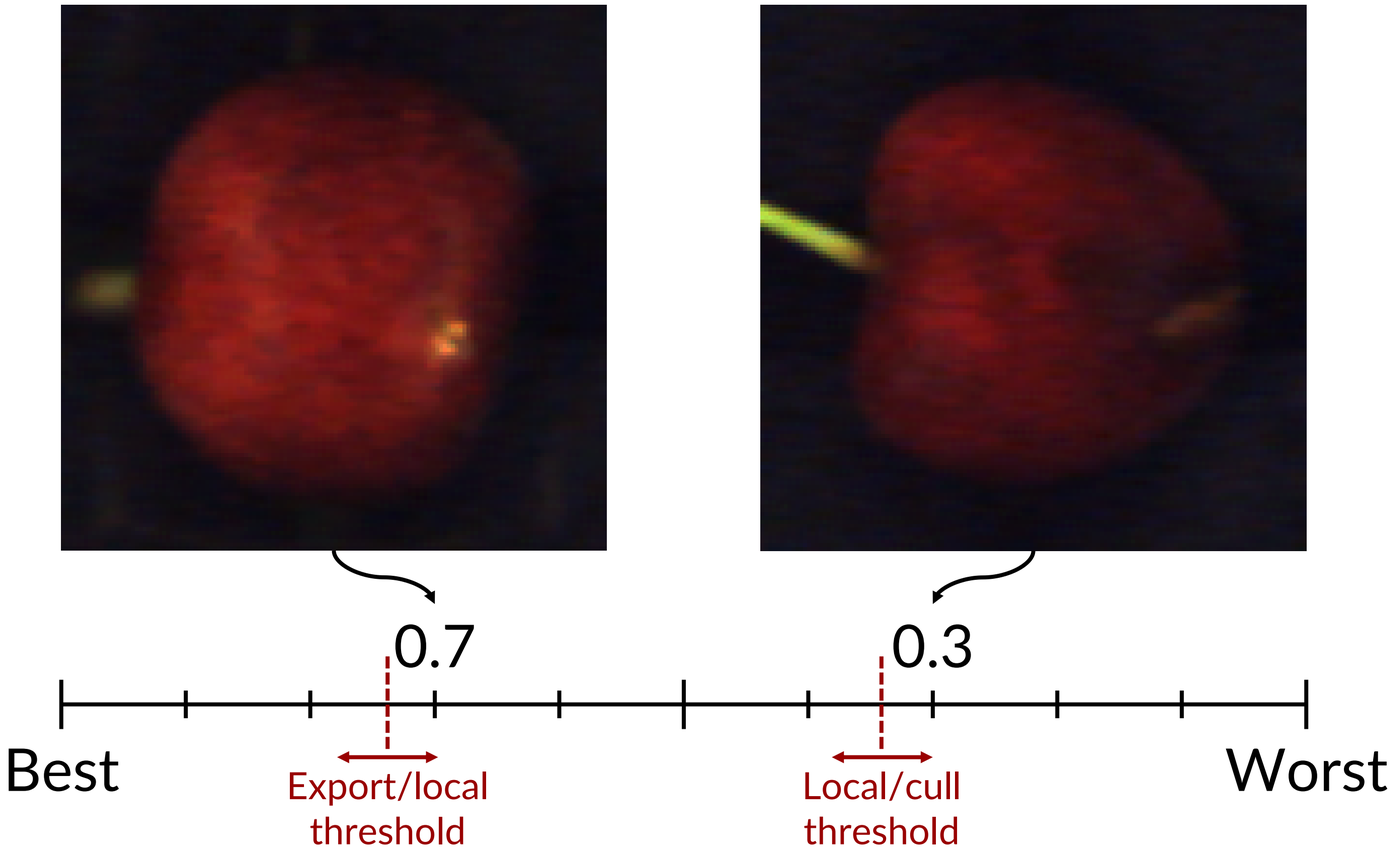
The Real Problem

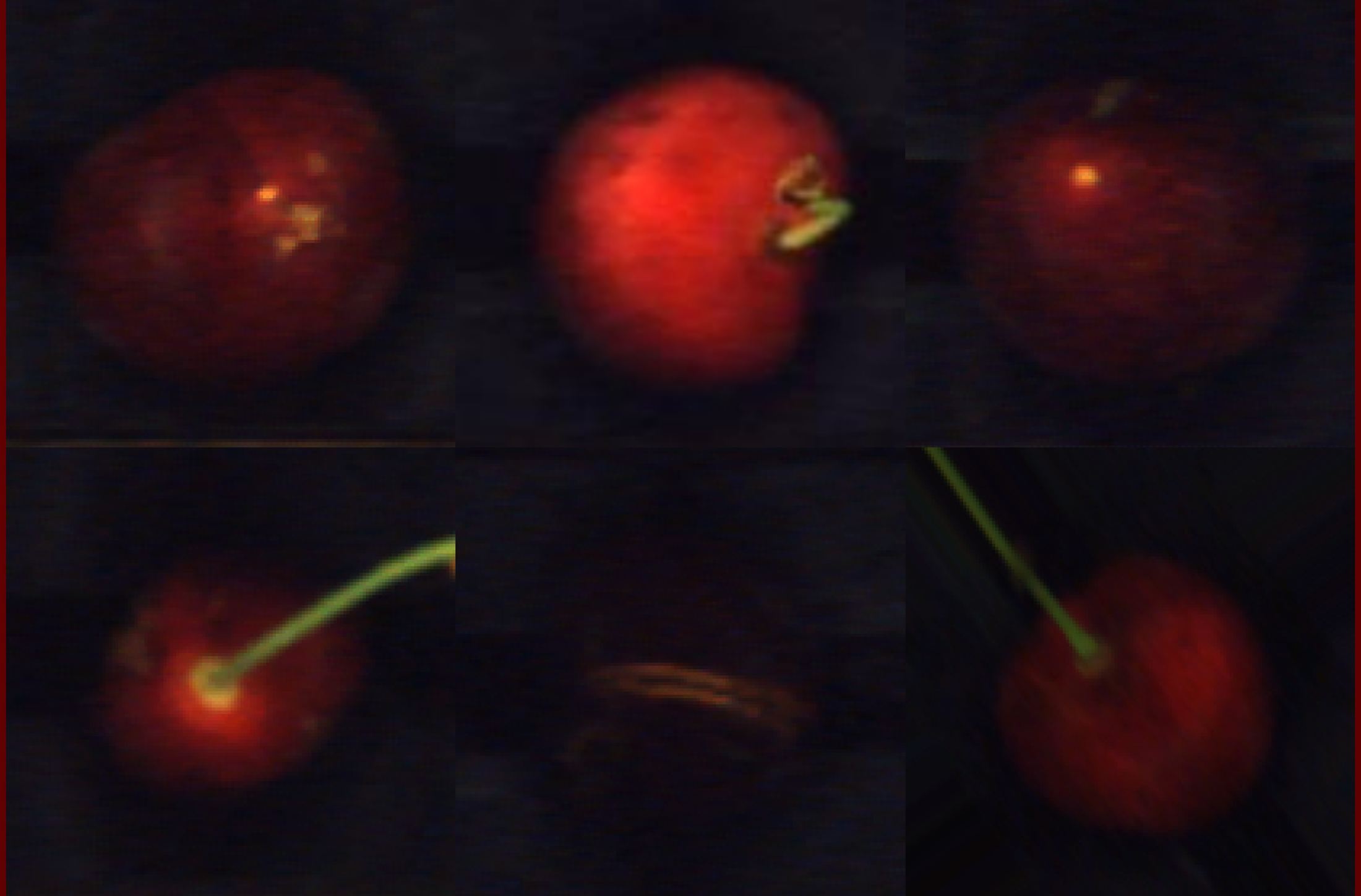
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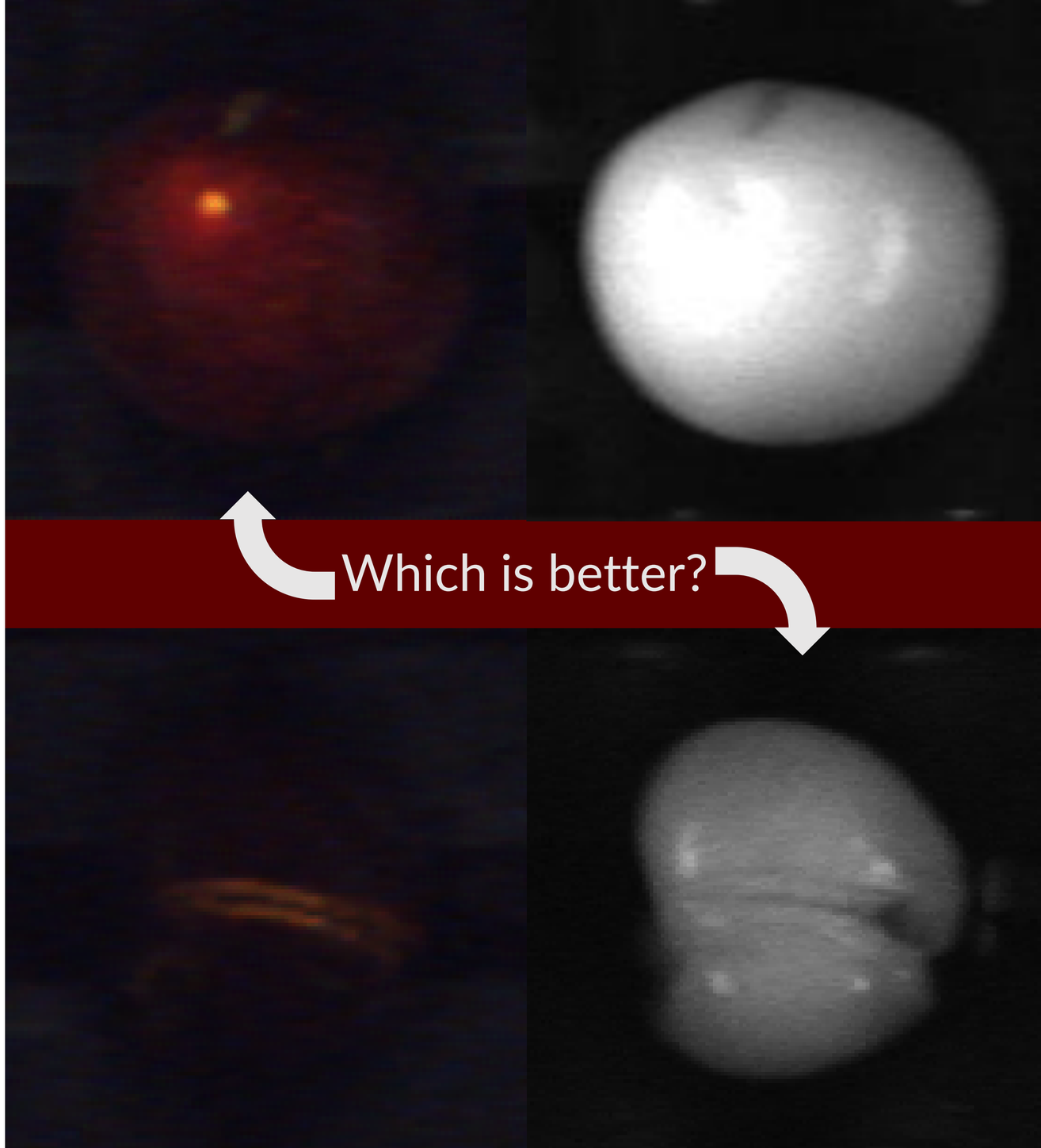
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Cannot use fixed classes

- Grade standards constantly change
- Traditional classification methods unsuitable

Continuous scale

- Train a model to predict ‘quality values’
- Just re-label images with quality values…? Difficult!
- Labelling on a consistent, continuous scale isn’t going to work.



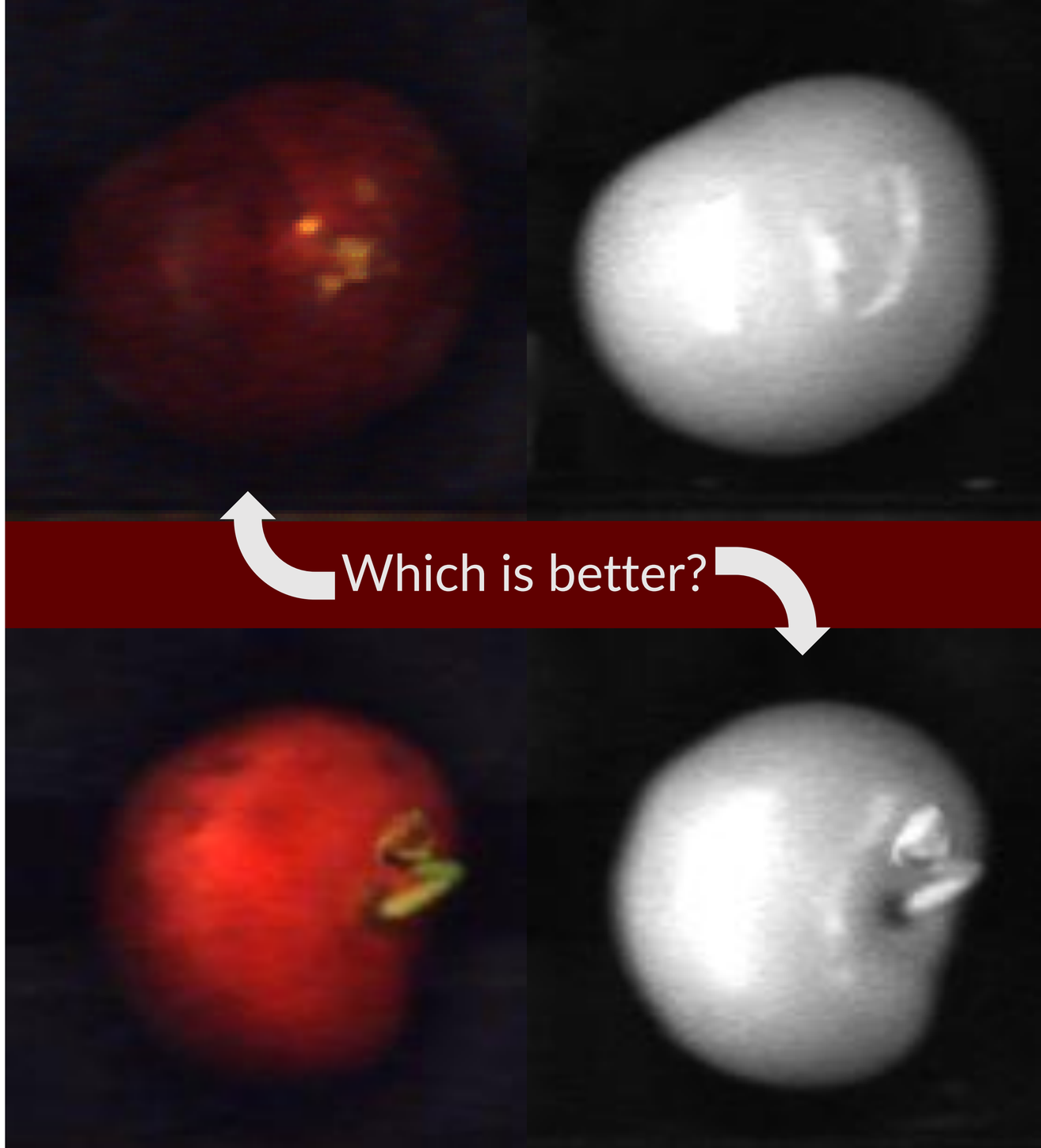
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The Real Problem

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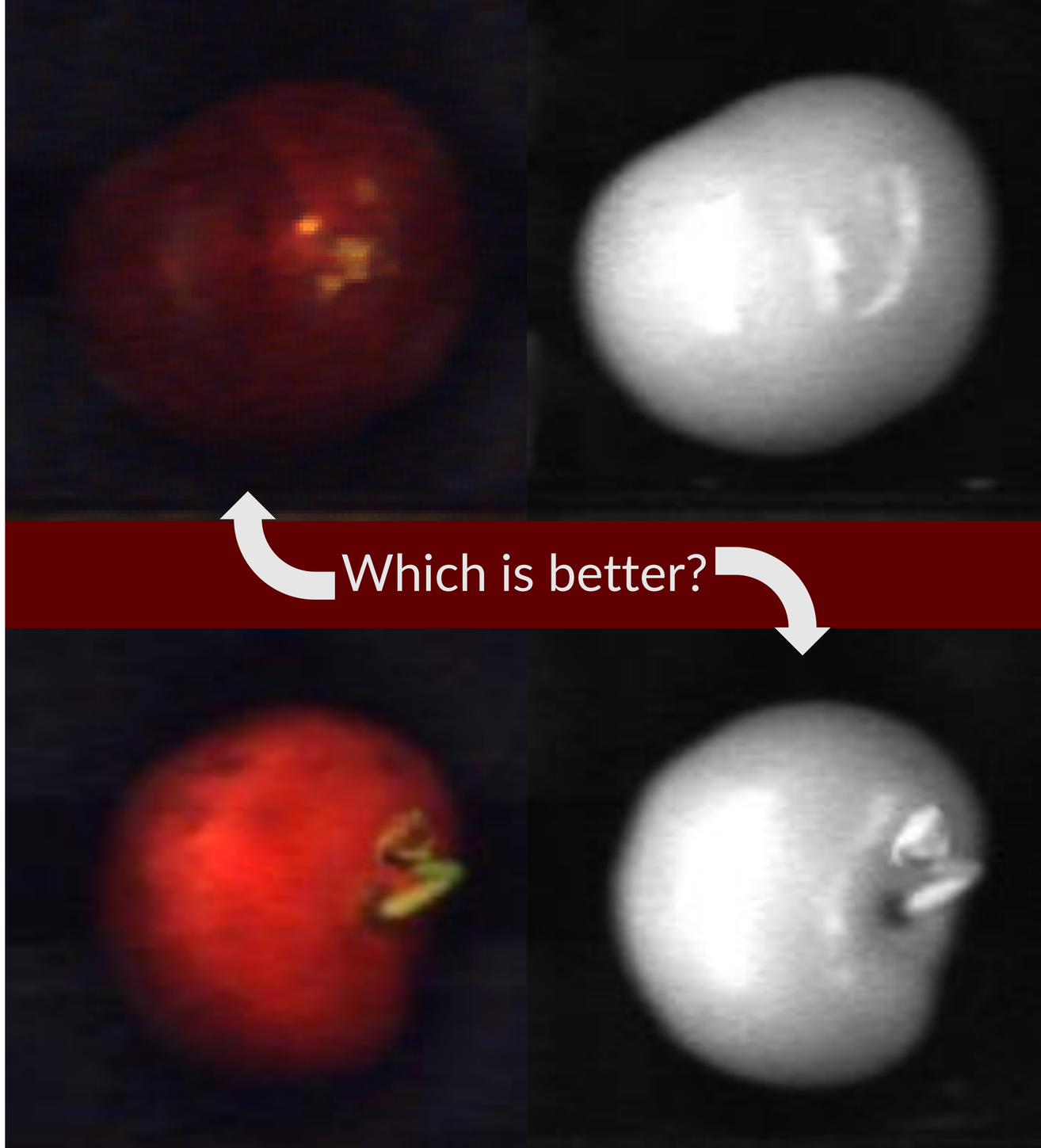
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Continuous scale

- Train a model to predict ‘quality values’
- Just re-label images with quality values…? Difficult!
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Pairwise comparisons

- Much easier for humans to do consistently
- Don’t need any idea of “absolute quality values”

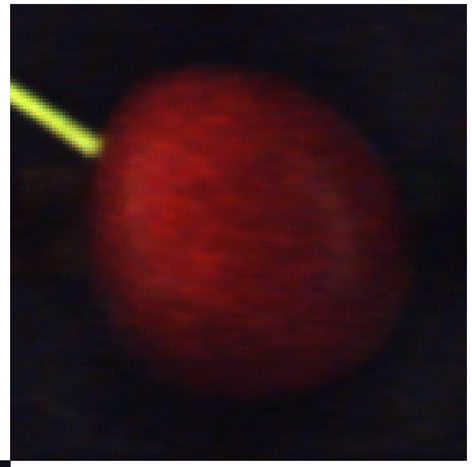


Labels

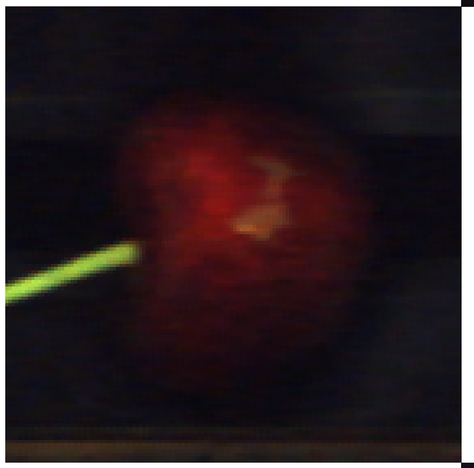
Initial labels

- Export, local, and cull
- Correspond to the grade standards at a **single point in time**.

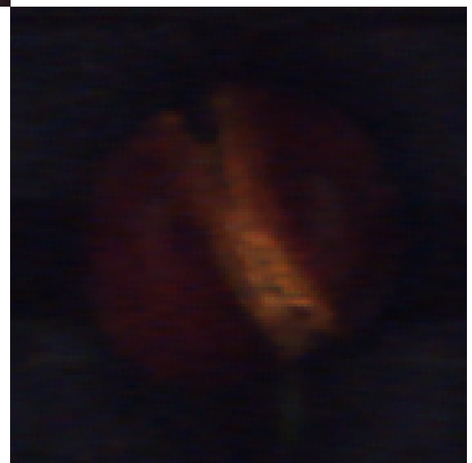
Export



Local



Cull



Labels

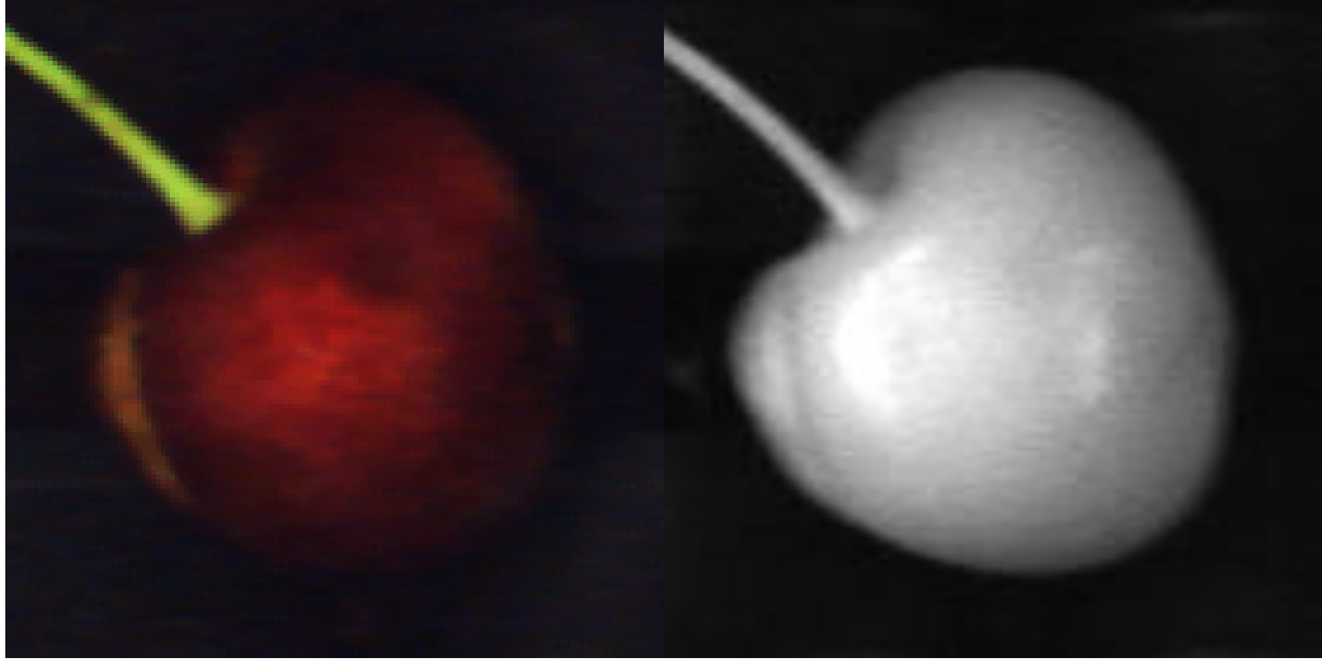
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Pairwise comparisons

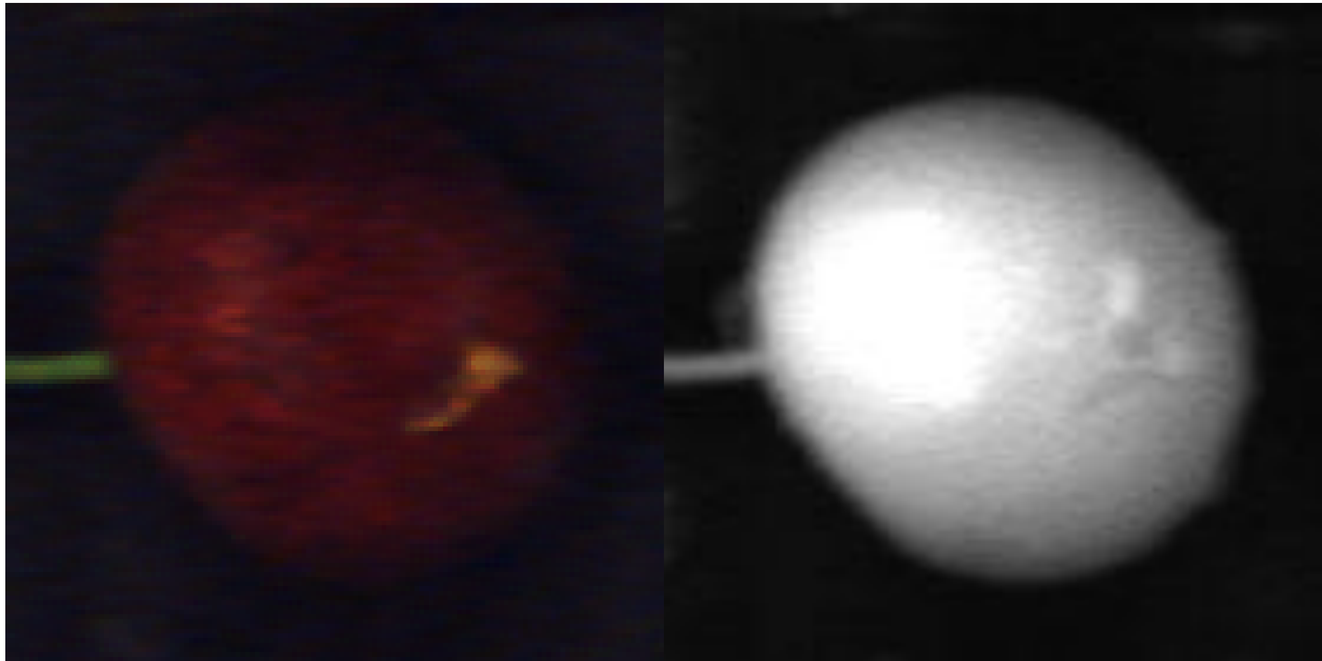
- Between-class

Cull



VS

Local



Labels

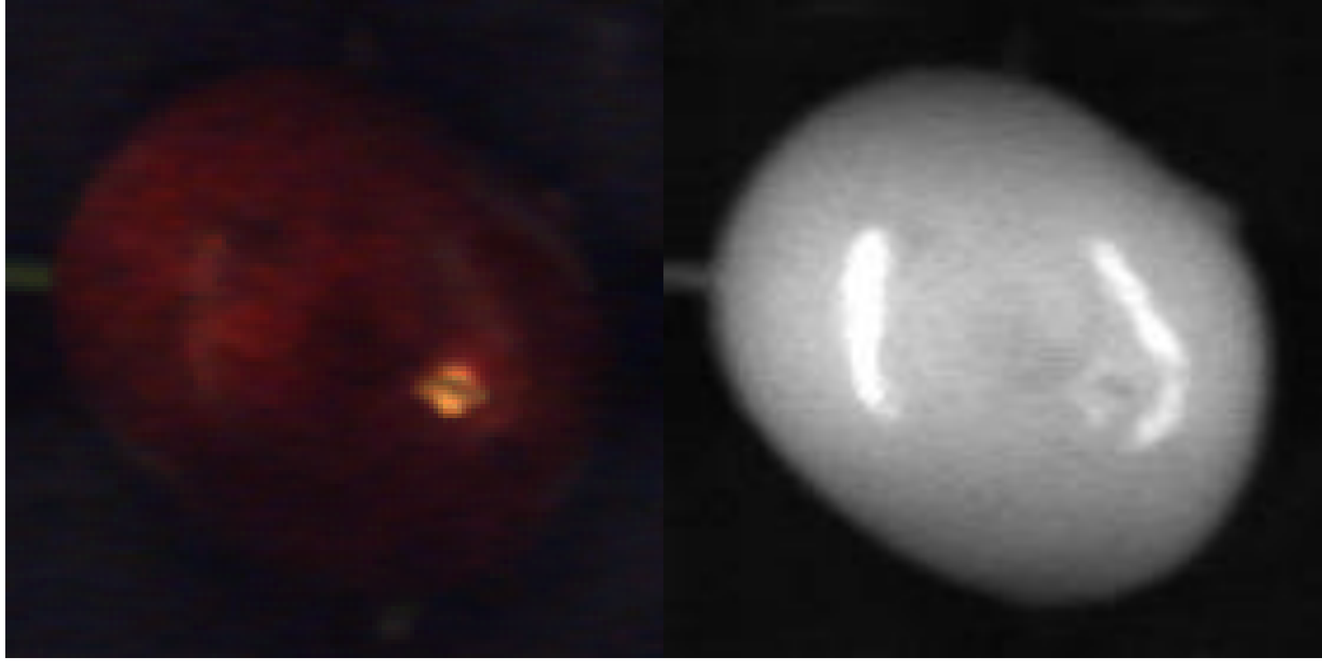
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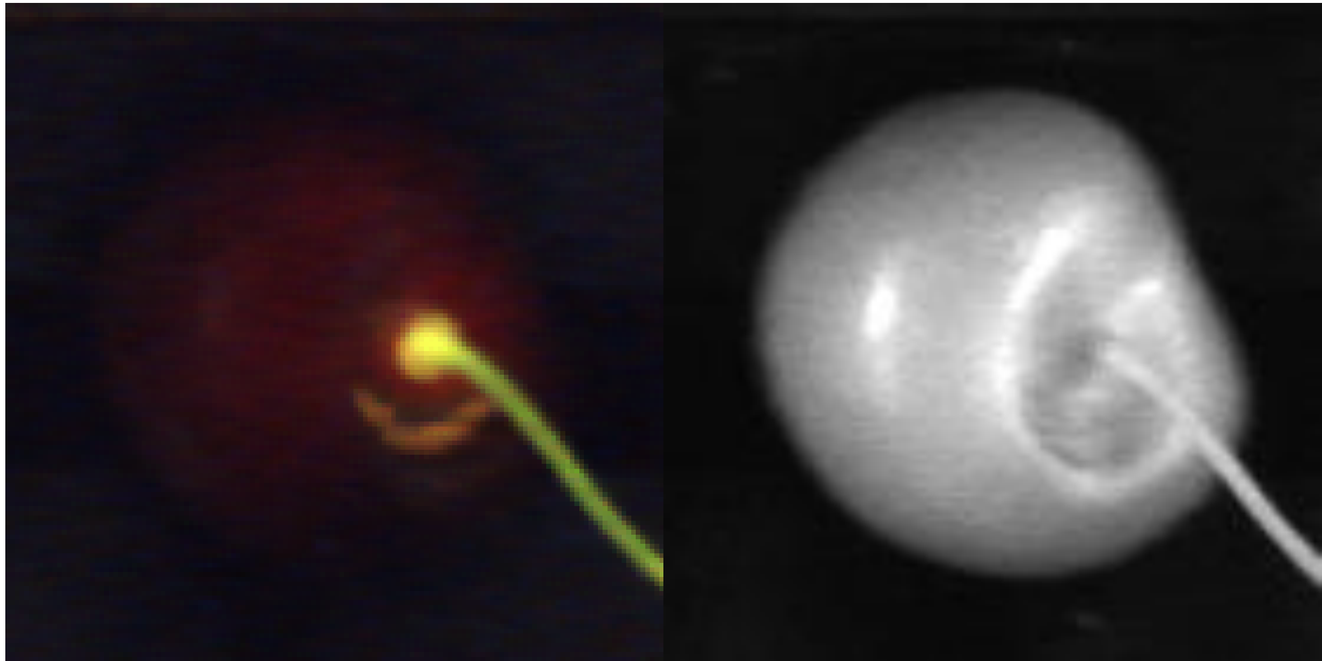
- Between-class
- Within-class
  - Better
  - Worse
  - Equal

Local



VS

Local



The Solution

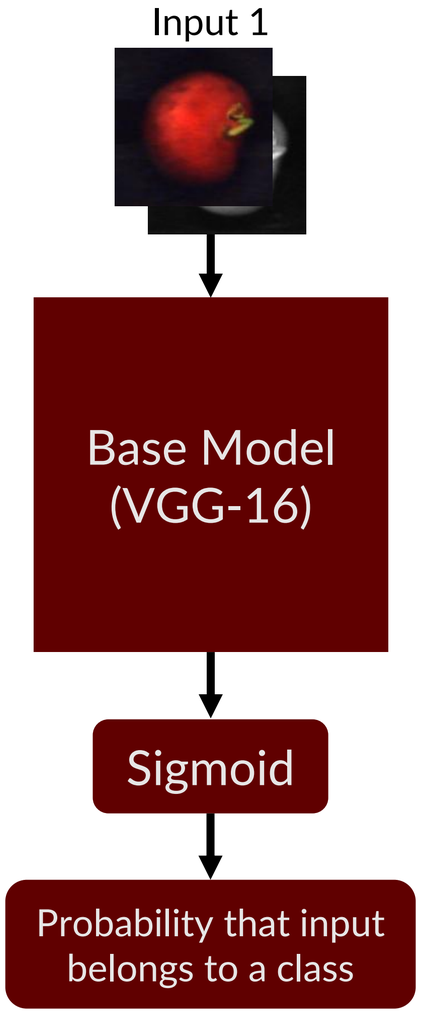
The Solution

Siamese Networks

Siamese Networks

Fundamental Architecture

- Siamese Networks



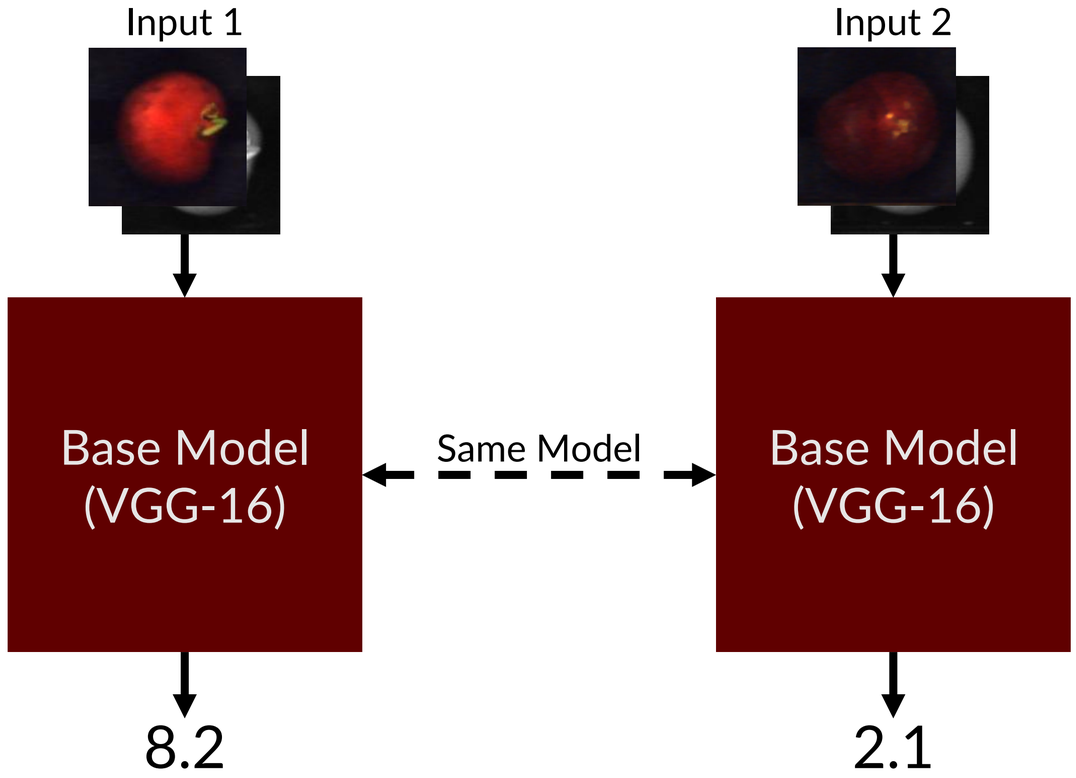
VGG-16 (Slightly modified)

Layer	Type	Output Shape
-	Input Layer	(140, 140, 4)
1	Convolutional (ReLU, 3x3, 64 filters)	(140, 140, 64)
2	Convolutional (ReLU, 3x3, 64 filters)	(140, 140, 64)
-	Max Pooling (2x2)	(70, 70, 64)
-	Batch Normalization	(70, 70, 64)
3	Convolutional (ReLU, 3x3, 128 filters)	(70, 70, 128)
4	Convolutional (ReLU, 3x3, 128 filters)	(70, 70, 128)
-	Max Pooling (2x2)	(35, 35, 128)
-	Batch Normalization	(35, 35, 128)
5	Convolutional (ReLU, 3x3, 256 filters)	(35, 35, 256)
6	Convolutional (ReLU, 3x3, 256 filters)	(35, 35, 256)
7	Convolutional (ReLU, 3x3, 256 filters)	(35, 35, 256)
-	Max Pooling (2x2)	(17, 17, 256)
8	Convolutional (ReLU, 3x3, 512 filters)	(17, 17, 512)
9	Convolutional (ReLU, 3x3, 512 filters)	(17, 17, 512)
10	Convolutional (ReLU, 3x3, 512 filters)	(17, 17, 512)
-	Max Pooling (2x2)	(8, 8, 512)
11	Convolutional (ReLU, 3x3, 512 filters)	(8, 8, 512)
12	Convolutional (ReLU, 3x3, 512 filters)	(8, 8, 512)
13	Convolutional (ReLU, 3x3, 512 filters)	(8, 8, 512)
-	Max Pooling (2x2)	(4, 4, 512)
-	Batch Normalization	(4, 4, 512)
-	Flatten	8192
14	Fully Connected (ReLU)	4096
15	Fully Connected (ReLU)	4096
16	Fully Connected (Linear)	1

Siamese Networks

Fundamental Architecture

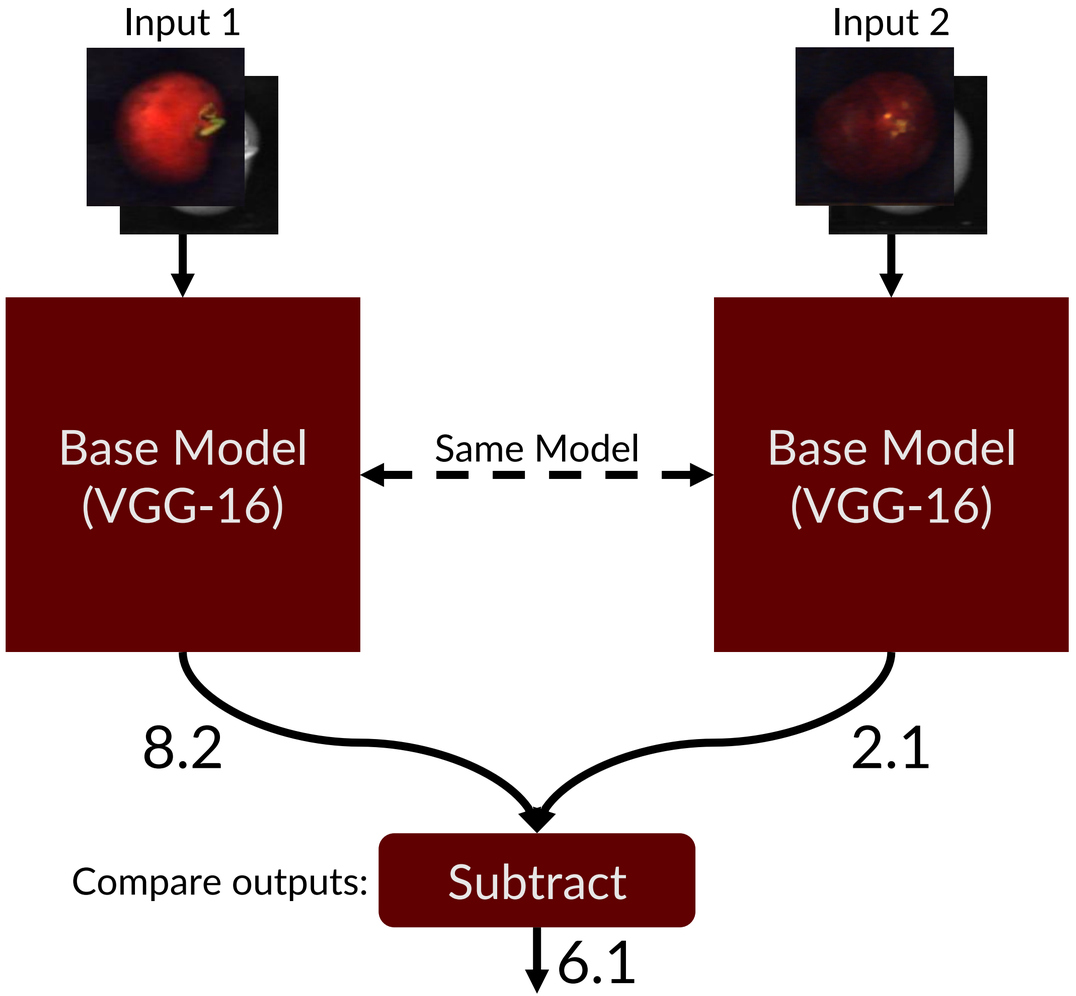
- Siamese Networks



Siamese Networks

Fundamental Architecture

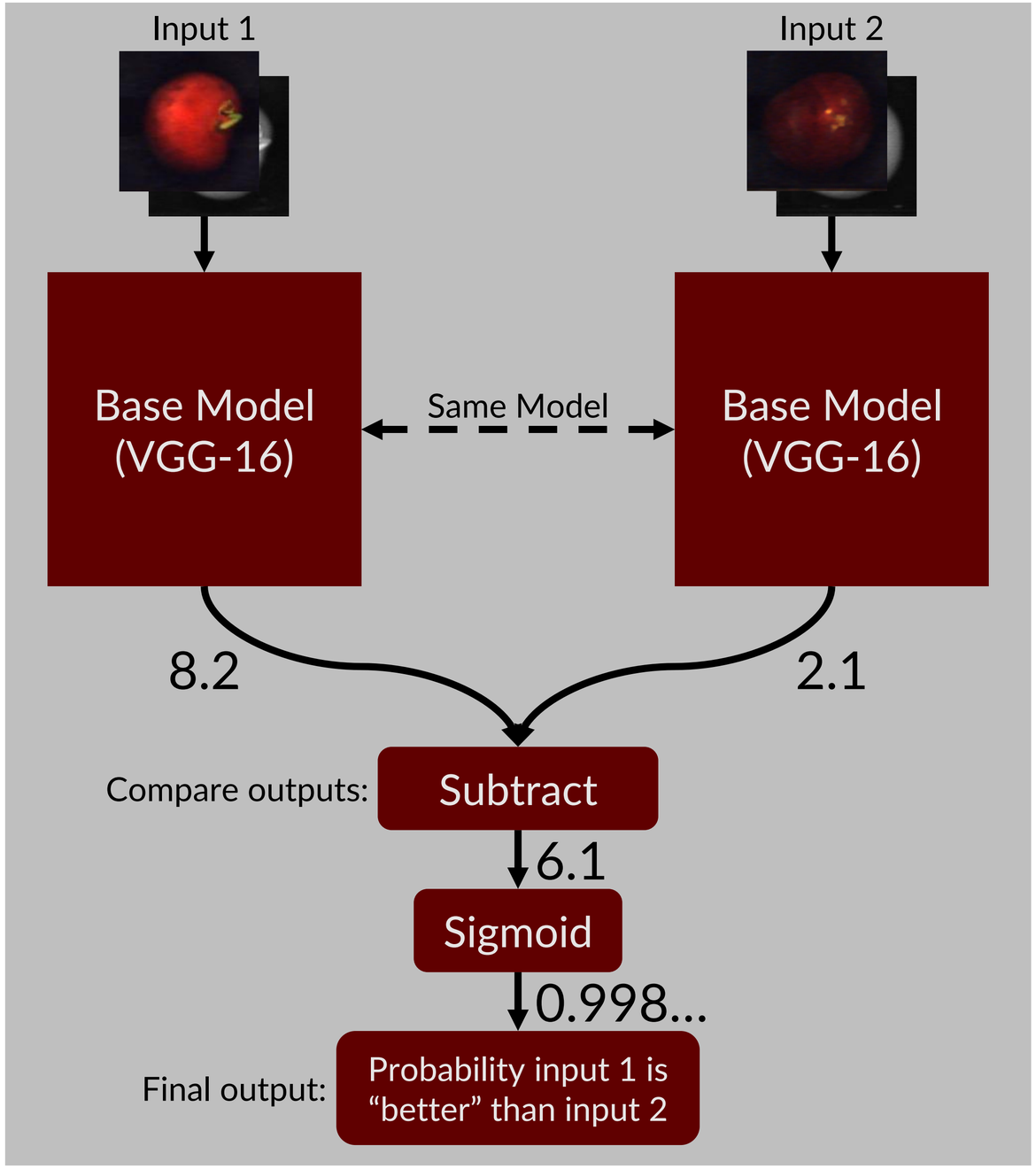
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Siamese Networks

Fundamental Architecture

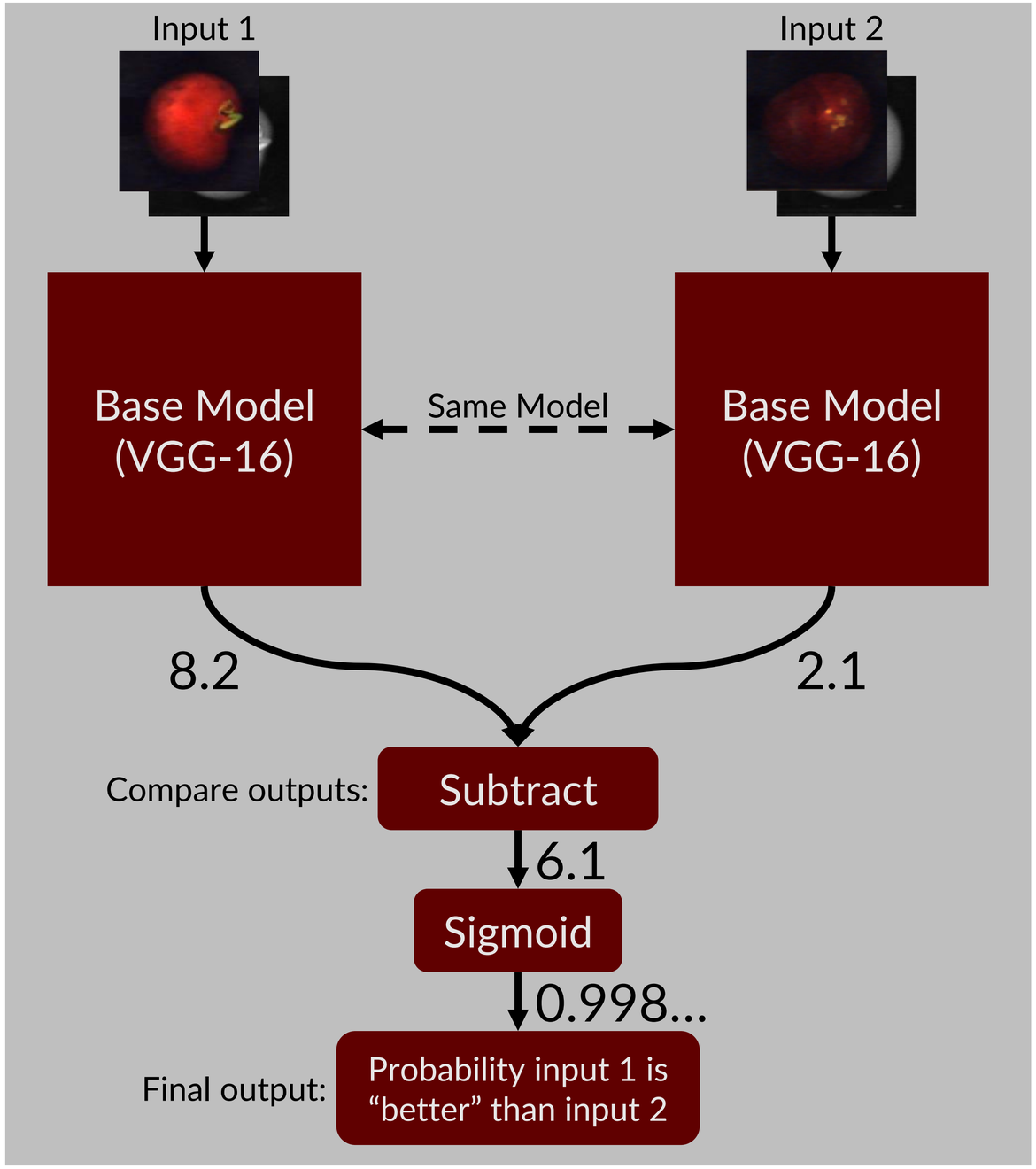
- Siamese Networks



Siamese Networks

Fundamental Architecture

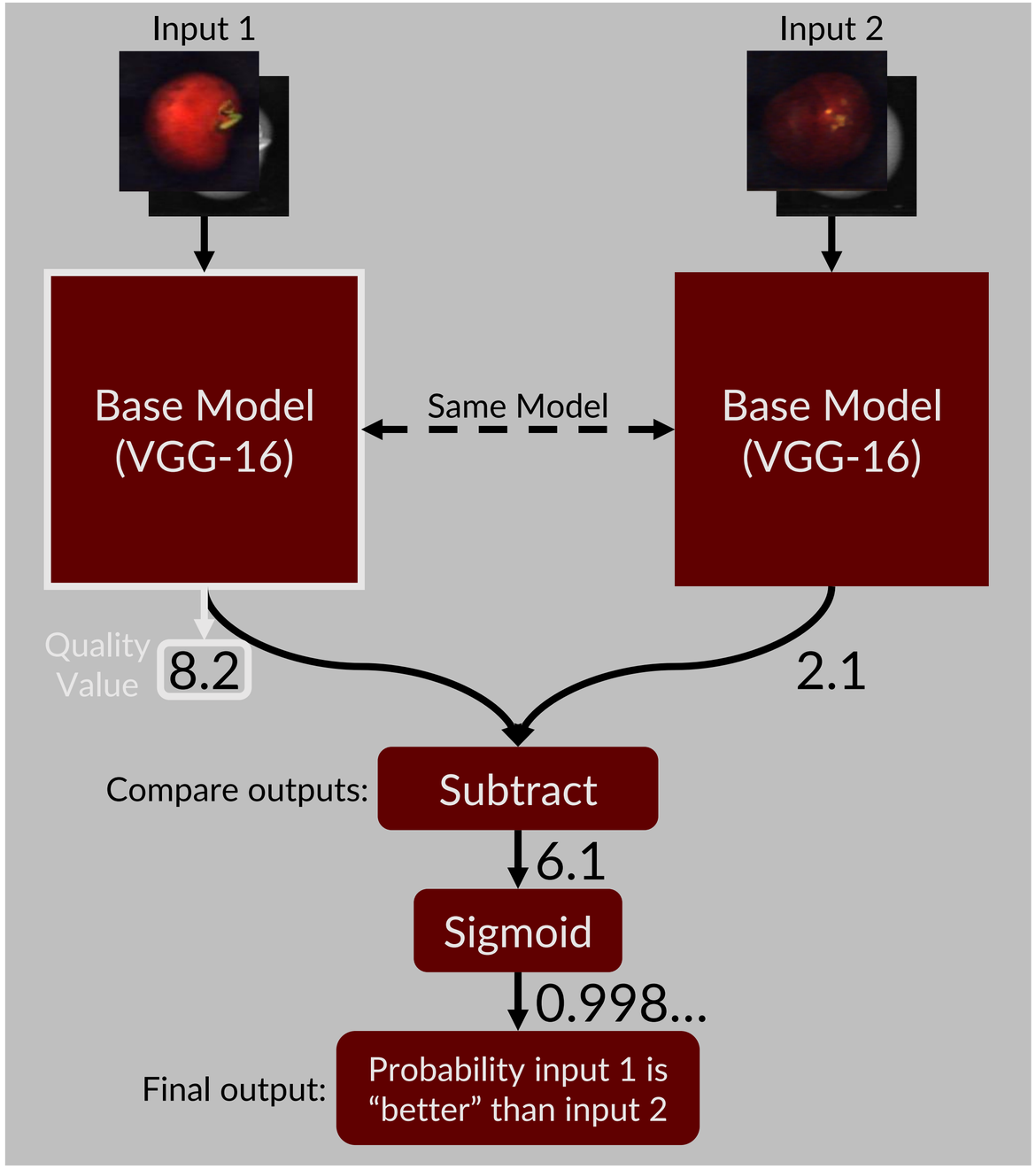
- Siamese Networks
- How to transform into absolute values?



Siamese Networks

Fundamental Architecture

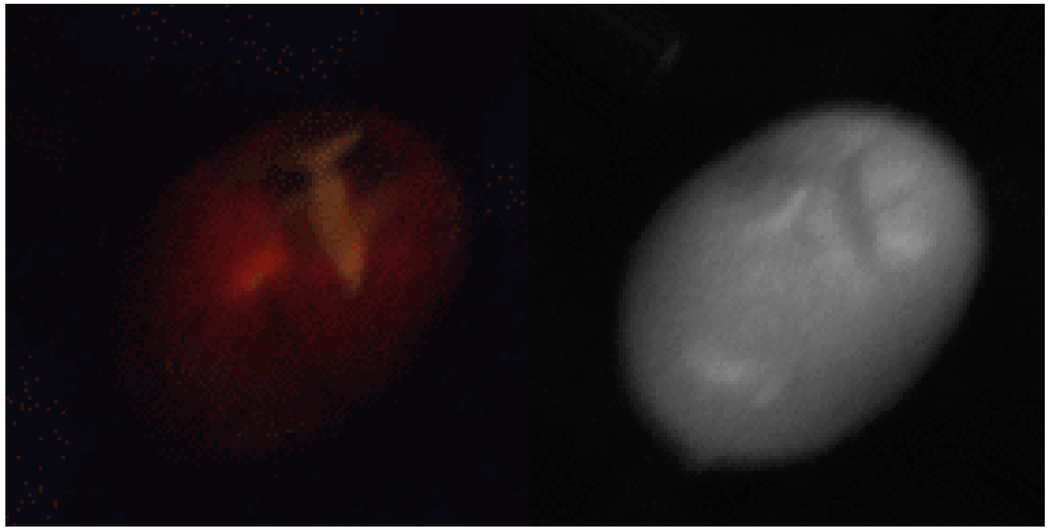
- Siamese Networks
- How to transform into absolute values?
    - Just extract the base model



Augmentation

Process

- Online augmentation each epoch



Augmentation Pipeline

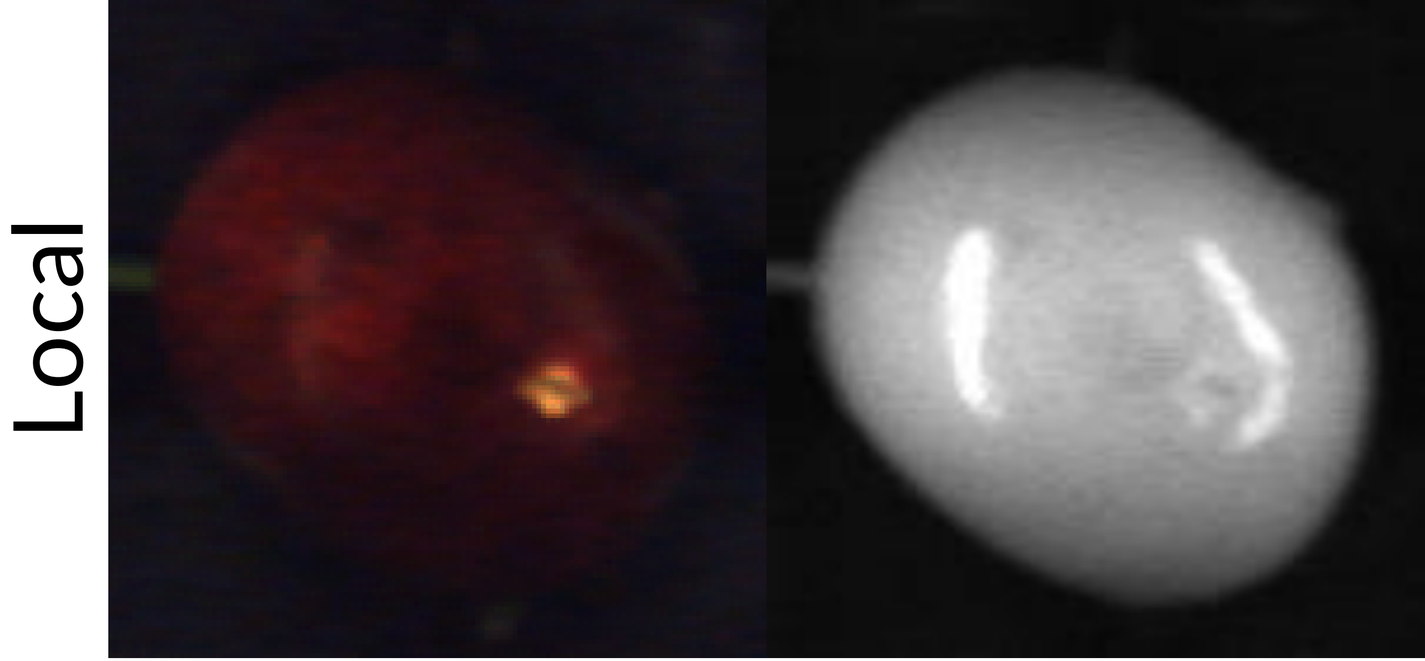
Augmentation	Probability	Value Range (Randomly Sampled)
Pixel-wise multiply (brightness)	1	(0.5, 1.5)
Horizontal flip	0.5	-
Vertical flip	0.5	-
Horizontal scale	1	(0.8, 1.2) proportion
Vertical scale	1	(0.8, 1.2) proportion
Horizontal translation	1	(-0.1, 0.1) proportion
Vertical translation	1	(-0.1, 0.1) proportion
Shear	1	(-8, 8) degrees
Rotation	1	(0, 360) degrees
Gaussian blur	0.25	(0.0, 3.0) sigma

Evaluation and Results

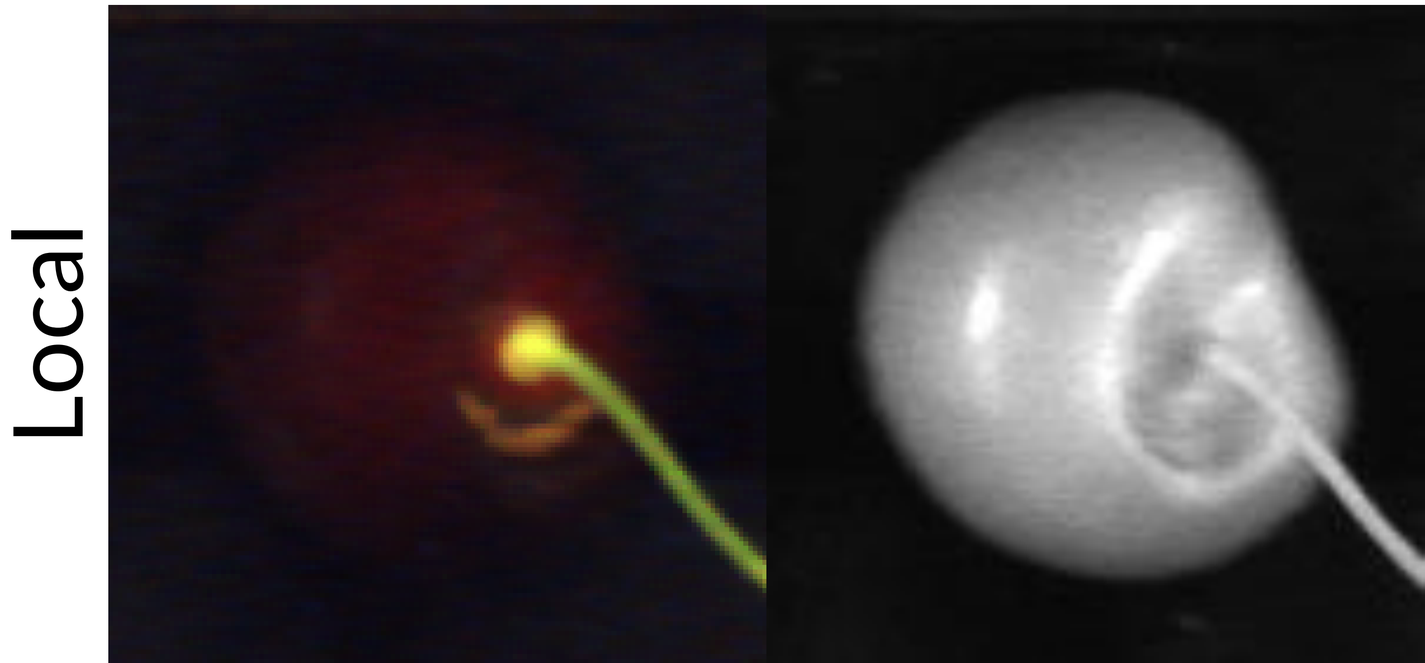
Evaluation Metrics

Pairwise accuracy

- Within-class
    - Only better/worse comparisons evaluated
- Between-class



VS



Results

	Within-class pairwise accuracy	Between-class pairwise accuracy	Three-class accuracy
Human self-agreement	93.1%	-	-
Standard classifier	60.4%	98.4%	92.6%
Siamese network	88.1%	98.0%	87.9%

Results

Human self-agreement

- Agreement between two rounds of test set labelling
- Rounds performed a week apart

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Standard Classifier

- Base model (VGG-16)
- Trained on initial, **fixed** class labels (export, local, cull)

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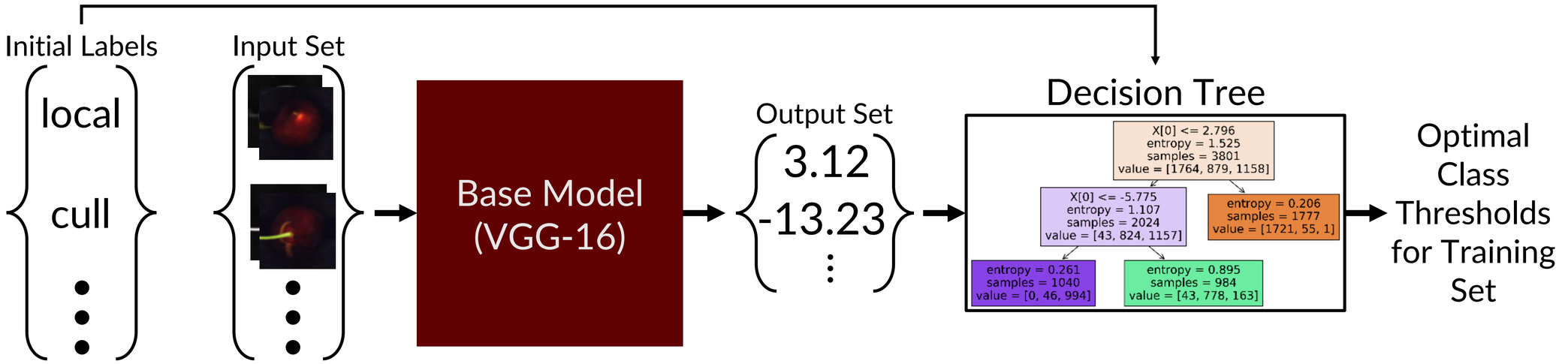
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Standard classifier	60.4%	98.4%	92.6%
Siamese network	88.1%	98.0%	87.9%



Results

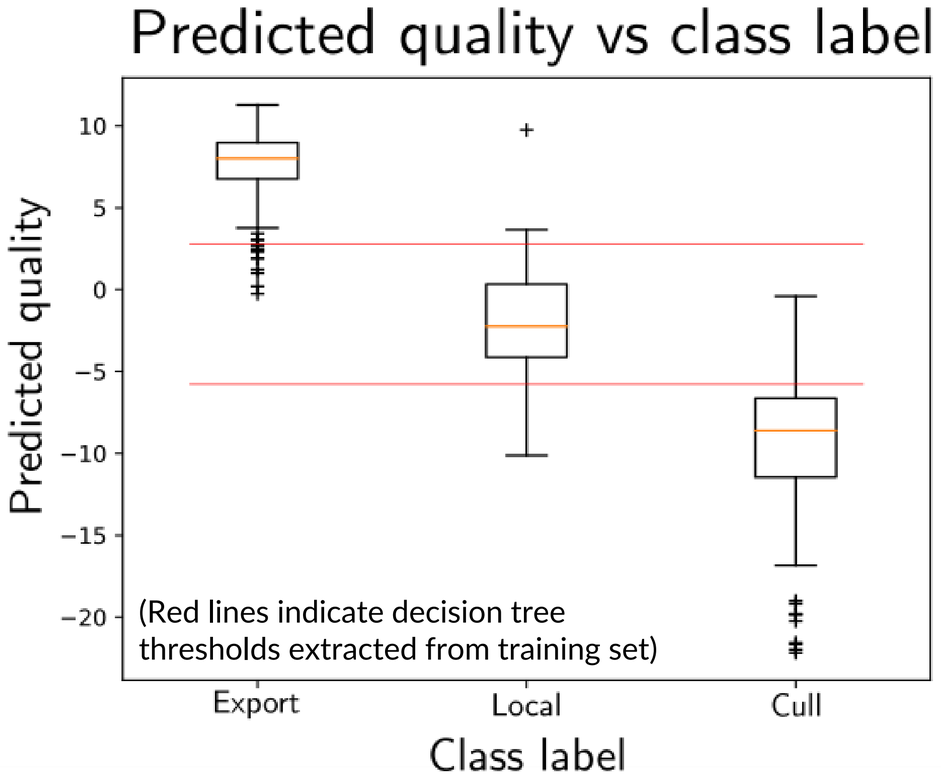
Human self-agreement

- Agreement between two rounds of test set labelling
- Rounds performed a week apart

Standard Classifier

- Base model (VGG-16)
- Trained on initial, **fixed** class labels (export, local, cull)

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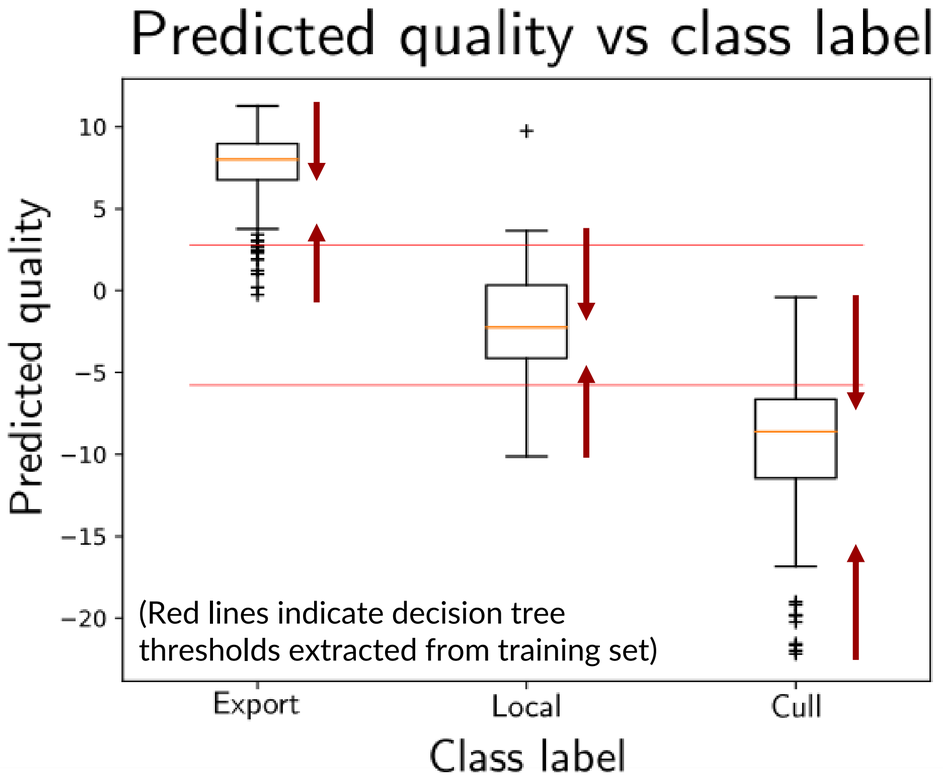
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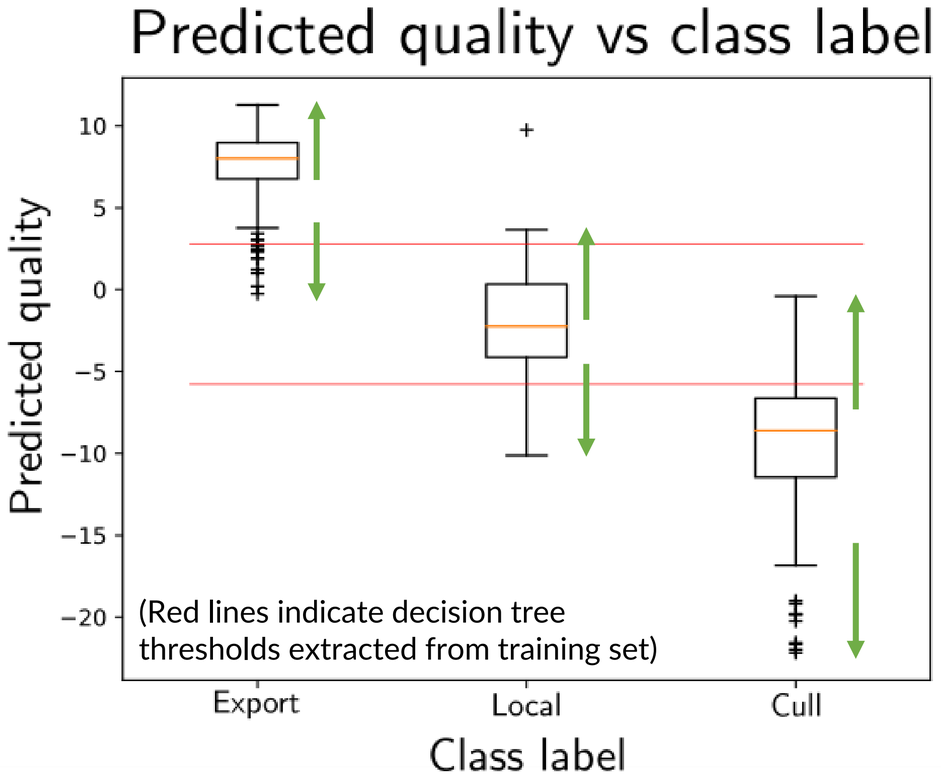
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Human self-agreement

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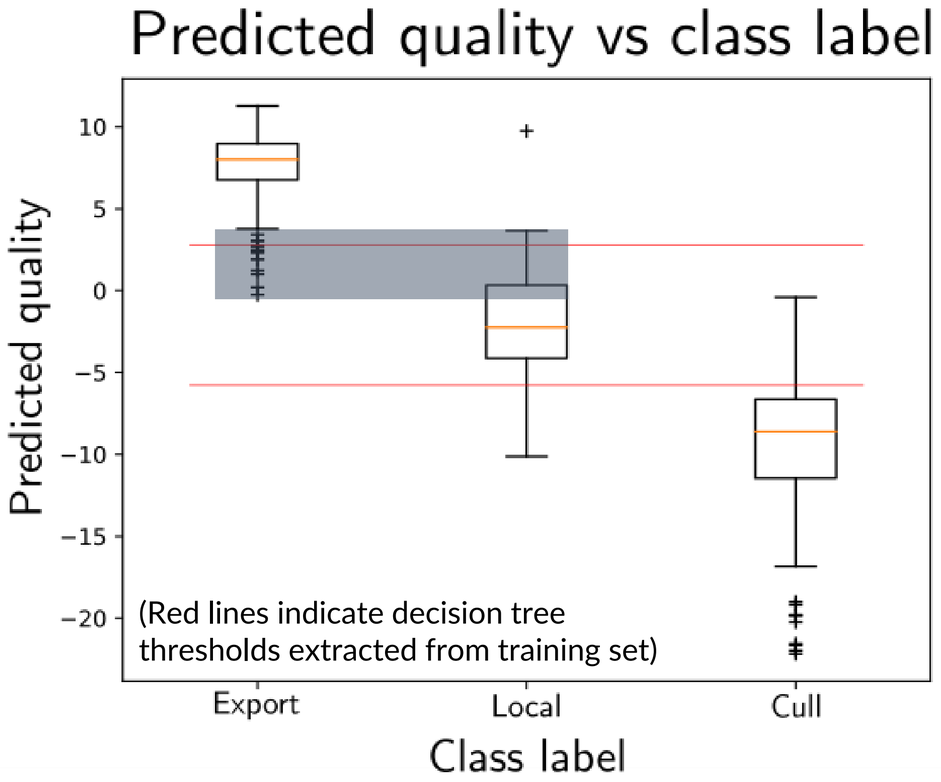
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Predicted quality distribution

- Minimal overlap between export and local
- More overlap between local and cull

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Results

Human self-agreement

- Agreement between two rounds of test set labelling
- Rounds performed a week apart

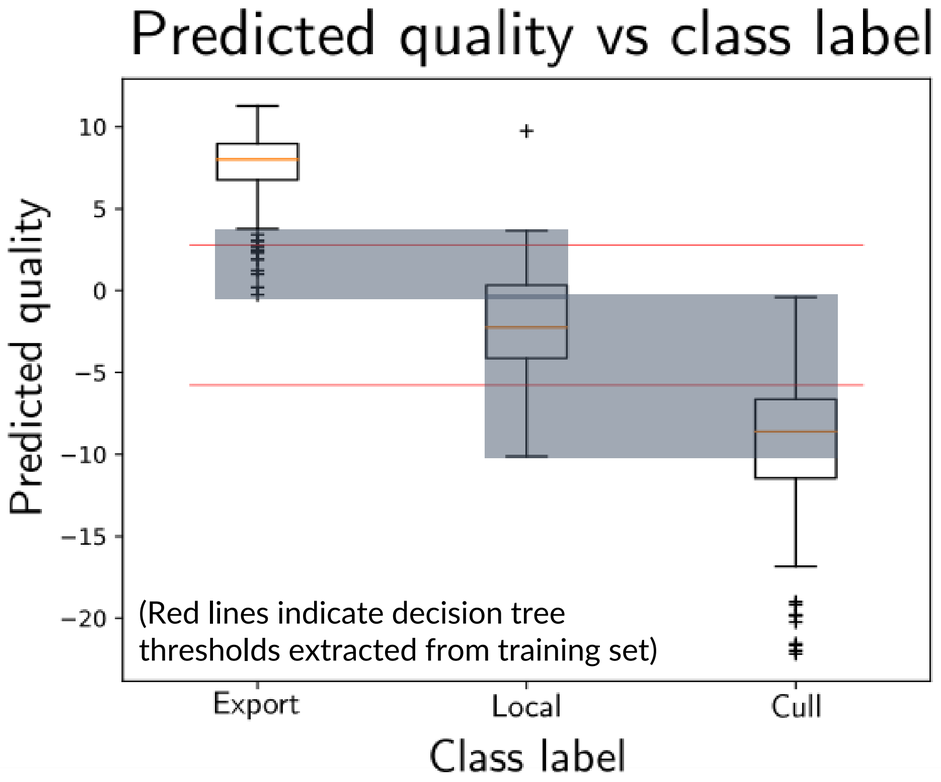
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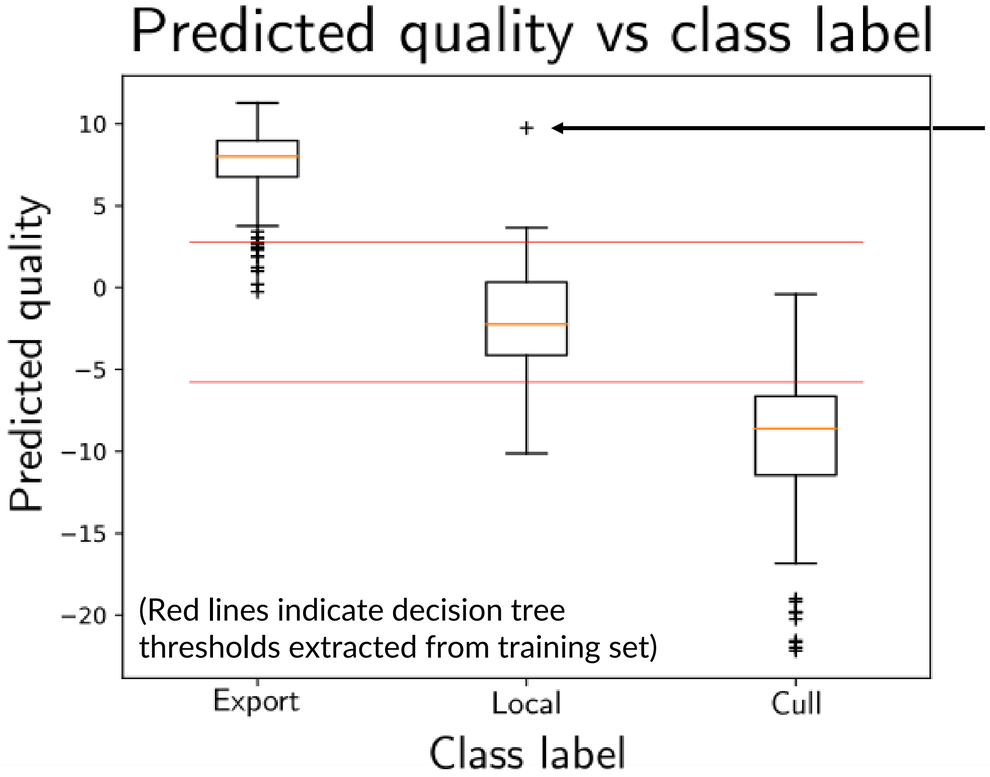
Standard Classifier

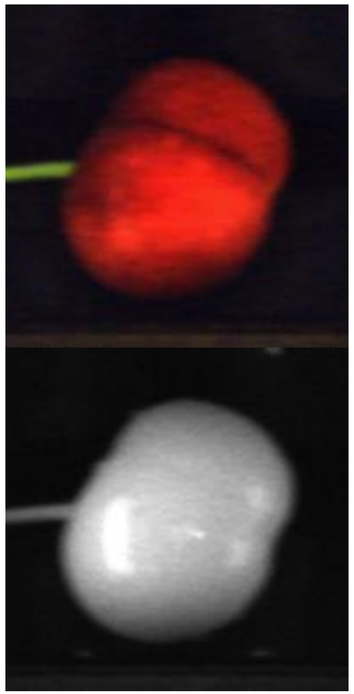
- Base model (VGG-16)
- Trained on initial, **fixed** class labels (export, local, cull)

Predicted quality distribution

- Minimal overlap between export and local
- More overlap between local and cull
- Outlier?

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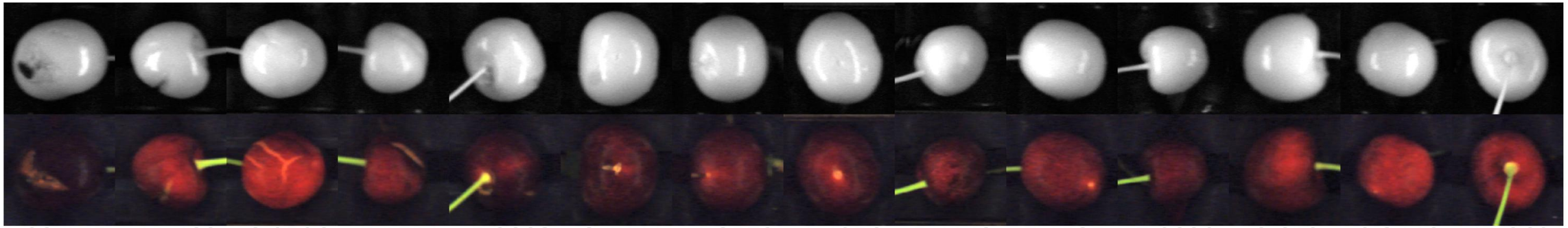




"Deformed"

Results

Every 38th image as ordered by the base model

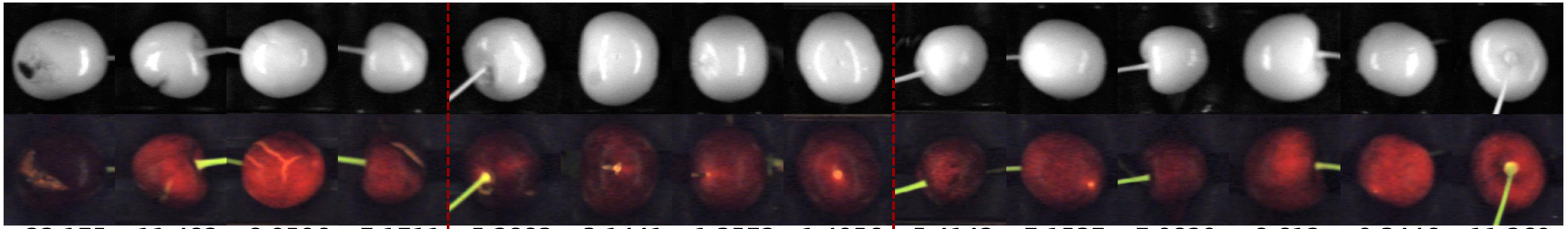


Ordering

- Correctly giving bad cherries low scores and good cherries high ones
- Smoothly change in quality

Results

Every 38th image as ordered by the base model



Ordering

- Correctly giving bad cherries low scores and good cherries high ones
- Smoothly change in quality

Final Words

Looking forward

- Further testing in the production environment
- Semi-supervised learning
- Adaptive learning

Beyond Cherries

Wider applications

- Essentially any situation with continuous attributes
    - Estimating photo quality/attractiveness/beauty
    - Gradable adjectives
    - …
- Not just limited to image tasks
    - Essay grading systems
    - Sentiment analysis
    - …

Thank you!

ANY QUESTIONS?

Beyond Cherries

Wider applications

- Essentially any situation with continuous attributes
  - Estimating photo quality/attractiveness/beauty
  - Gradable adjectives
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- Not just limited to image tasks
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